Swap Curve Construction and Optimization of the Tension Parameter: An Energy-Partition Formulation

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Abstract

Polynomial splines have long been used for the construction of swap curves. They suffer, however, from excessive convexity, a lack of locality under perturbation of the input rates, and the possibility of negative implied forwards. Tension splines remedy these defects through a single shape parameter, but the choice of that parameter has remained largely heuristic. In this paper we place the tension spline on its proper physical footing as the equilibrium shape of a thin elastic beam held under longitudinal tension, and we derive in closed form the two energies stored by such a beam: a *flexural* (bending) energy and an *axial* (tension) energy. We show that the curvature "smoothness" norm used in the literature *is* the flexural energy, and that the first-derivative norm *is* the axial energy, so that the two are the complementary halves of one elastic energy rather than independent quantities. We then propose selecting the tension parameter by the *bending-energy fraction* $\varphi(\sigma) = E_{\text{bend}}/(E_{\text{bend}} + E_{\text{tens}})$, a dimensionless quantity that runs monotonically from one (cubic) to zero (linear). The neutral choice $\varphi(\sigma^\star) = \frac{1}{2}$ corresponds to equipartition of elastic energy; the target $\varphi^\star$ provides a single, transparent dial through which a desk may express its trade-off between smoothness and locality. The criterion is invariant under rescaling of the rate and time axes, removing the unit dependence of earlier energy-based schemes. On the U.S. Treasury par-yield curve of 22-May-2026 the criterion yields a unique optimum ($\sigma^\star = 4.86$ at equipartition) and, relative to the natural cubic spline, reduces the hedge-locality radius of the benchmark 5-year node from 6.7 years to 0.8 year — a factor of eight — while suppressing the spurious long-end forward hump that the cubic manufactures from a flat 20–30-year segment. Across 1,064 weekly U.S. Treasury curves spanning 2006–2026, the selected tension tracks the rate regime smoothly without jumps, and the predicted inverse relationship between tension and hedge-locality radius holds out of sample with a log-correlation of -0.99.

Keywords: yield curve construction; term-structure interpolation; tension spline; energy-partition criterion; hedge locality; instantaneous forward curve; multi-curve SOFR/OIS; variational method; Euler–Bernoulli beam.

JEL Classification: C63; G12; E43; C61; G17.

I. Introduction

A risk-free term structure is fundamental to the pricing of fixed-income securities and to the risk management of the same. The term structure is required to be a continuous function of time, with additional requirements on the continuity of its first and second derivatives. Given these constraints, computational procedures are needed to estimate the structure, and for those procedures the Treasury rates alone are not enough: to reflect the market a combination of instruments — cash deposits, futures, swap rates — is used to estimate the curve.

Finding the right set of instruments is only half the problem. As Hagan and West [8] articulate, there is no single best way to construct a term structure from a set of rates. It is commonly desired that the derived curve be smooth, that the interpolation be *local*, that the implied forwards be both smooth and stable, and that hedges be as local as possible. Smoothness means the absence of kinks and spurious wiggles; market fidelity means the curve reproduces known market rates exactly; locality means that a perturbation of one input propagates to a bounded neighbourhood of the curve, which is precisely what permits a fixed-income portfolio to be hedged with a small, well-localised set of instruments.

Parametric methods such as Nelson–Siegel produce a good overall shape but leave much to be desired in the accuracy of fit. Cubic regression splines, first applied to the term structure by McCulloch [11], give the smoothest interpolant and have been shown to be at least as satisfactory as competing methods. Cubic splines are not, however, free of defects: they fail to guarantee monotonicity or convexity of the data [1], they are non-local, and for rate curves they can produce negative forwards. Tanggaard [12] introduced non-parametric smoothing of the yield curve through a penalised least-squares estimator; Anderson [1] extended this idea, modifying the penalty term to obtain hyperbolic tension splines. Kamada and Enkhbat [10] present the tension spline as a generalisation that interpolates between the cubic (minimiser of the squared second derivative) and the linear (minimiser of the squared first derivative) splines, and so provides a convenient middle ground that preserves shape [7]. Costantini, Kvasov and Manni [7] treat the hyperbolic tension spline as the solution of a differential multipoint boundary-value problem; Renka [9] automates the choice of the tension factor subject to monotonicity and convexity constraints.

The practical relevance of these defects is underscored by official practice. The U.S. Treasury, which publishes the most widely referenced government curve in the world, abandoned its quasi-cubic Hermite spline in December 2021 and now estimates its daily par-yield curve with a monotone convex spline [16], precisely to suppress the oscillation and non-locality of the cubic. The tension spline occupies exactly the continuum between these two regimes: it relaxes to the cubic at zero tension and tightens towards the shape-preserving, local, piecewise-linear limit as the tension grows, with a single parameter governing the position along that continuum.

We make two methodological observations about this body of work. First, automated selection of the tension factor, while elegant, does not by itself respond to a practitioner's risk and hedging preferences; any defensible selection rule should expose those preferences as an explicit parameter rather than hide them inside a fixed constraint set. Second — and this is the technical heart of the paper — the "elastic beam" analogy that motivates the tension spline can be made fully rigorous. The tension spline is not merely *like* a beam; it *is* the equilibrium shape of a thin elastic beam under longitudinal tension, and the norms that the literature uses to score a curve are the literal flexural and axial energies of that beam. Recognising this collapses the two seemingly independent "smoothness" and "length" penalties into the two halves of a single stored energy, and it suggests a natural, dimensionless, range-independent rule for selecting the tension. This paper revises and substantially extends an earlier treatment by the author, recasting that elastic-beam heuristic on a rigorous variational footing and replacing the ad hoc selection rule with the dimensionless, scale-invariant criterion developed here.

A note on scope. For clarity the core construction is presented on a single curve; current market practice discounts on an overnight (OIS/SOFR) curve within a multi-curve framework [2, 3], and Section X demonstrates the method directly in that setting. The interpolation theory developed below is agnostic to that choice: it acts downstream of the bootstrap and applies unchanged to any single curve, whether a projection curve or a discount curve. We also interpolate in the space of instantaneous forward rates, following the emphasis of Hagan and West [8] on the importance of the interpolation variable; the forward space is the most demanding test of an interpolant because forwards expose, through the derivative, the oscillations that zero rates conceal.

The remainder of the paper is organised as follows. Section II reviews fixed-income pricing in relation to vanilla swaps. Section III reviews the natural cubic spline. Section IV establishes the tension spline as a tensioned elastic beam and derives its variational characterisation. Section V derives the flexural and axial energies in closed form and presents the energy-partition selection criterion. Section VI validates the criterion through forward positivity and hedge locality. Section VII extends the method to the instantaneous forward curve, Section VIII to per-segment non-uniform tension, and Section IX unifies the two. Section X embeds the method in a multi-curve SOFR/OIS construction. Section XI derives the analytic risk Jacobians and the stability of the dial. Section XII benchmarks the method against production interpolators and characterises $\varphi^\star$. Section XIII sets out the practical benefits, and Section XIV discusses results and future directions. Appendix A derives the non-uniform spline, Appendix B the closed-form energy integrals, and Appendix C the numerically stable evaluation.

II. Fixed-Income Pricing Basics

A money-market account evolves as $dB(t) = r_t\, B(t)\, dt$, so that $B(t) = \exp\big(\int_0^t r_s\, ds\big)$ and the stochastic discount factor is $D(t,T) = \exp\big(-\int_t^T r_s\, ds\big)$, where r_t is the instantaneous short rate. The zero-coupon bond price $P(t,T)$ is the risk-neutral expectation of $D(t,T)$ and equals $D(t,T)$ when the rate process is deterministic. With $\tau(t,T)$ the year fraction to maturity, the continuously compounded zero rate $R(t,T)$ satisfies

$$P(t,T) = e^{-R(t,T)\,\tau(t,T)}, \qquad R(t,T) = -\frac{\ln P(t,T)}{\tau(t,T)}. \tag{1}$$

Simply-compounded market rates (for example term SOFR) accrue on an Act/360 basis,

$$L(t,T) = \frac{1 - P(t,T)}{\tau(t,T)\, P(t,T)}. \tag{2}$$

The continuously compounded forward rate for $t \le T_1 \le T_2$ is

$$F(t,T_1,T_2) = \frac{1}{\tau(T_1,T_2)} \ln \frac{P(t,T_1)}{P(t,T_2)}, \tag{3}$$

Table 1: *U.S. Treasury CMT par-yield curve,* 22*-May-*2026 *(Federal Reserve H.15).*

Tenor	1M	3M	6M	1Y	2Y	3Y	5Y	7Y	10Y	20Y	30Y
Yield (%)	3.72	3.68	3.79	3.86	4.13	4.18	4.27	4.41	4.56	5.06	5.07

and for simple compounding

$$F(t,T_1,T_2) = \frac{1}{\tau(T_1,T_2)}\left(\frac{P(t,T_1)}{P(t,T_2)} - 1\right). \tag{4}$$

As $T_2 \to T_1$ the forward rate becomes the instantaneous forward $f(t,T) = -\partial_T \ln P(t,T)$, whence $P(t,T) = \exp\big(-\int_t^T f(t,u)\,du\big)$ and $F(t,T_1,T_2) = \frac{1}{T_2-T_1}\int_{T_1}^{T_2} f(t,u)\,du$.

A vanilla payer swap exchanges a fixed leg for a floating leg from a future start date. Its defining attributes — notional (used only for cash-flow calculation), fixed rate, coupon frequency, business-day convention, floating index, day-count, effective and maturity dates — are standard. A swap has zero present value at inception; the swap rate is the fixed rate that makes it so. Bootstrapping the curve from cash, futures and swaps proceeds, following Hagan [8], through the recursion for the discount factor implied by the n-th swap,

$$B(t,t_n) = \frac{1 - R_n\sum_{j=1}^{n-1}\theta_j B(t,t_j)}{1 + R_n\theta_n}, \qquad r_n = -\frac{1}{\tau_n}\ln B(t,t_n), \tag{5}$$

where R_n is the par swap rate, θ_j the coupon year fractions, and the unit in the numerator should strictly be the discount factor to the (two-business-day) effective date, here approximated by one. Short-end deposit and futures stubs are converted to continuously compounded spot rates through $f_{sl} = \theta_{sl}^{-1}\ln\big[(1 + r_l\theta_l)/(1 + r_s\theta_s)\big]$ and $(1 + F\theta_F) = e^{f\theta_f}$ respectively.

Applying (5), or in the illustrative case below simply reading the published constant-maturity yields, produces the curve nodes used throughout this paper.

For a current, fully reproducible illustration we take the U.S. Treasury par-yield curve of 22-May-2026 (Table 1), as published in the Federal Reserve H.15 release [17]. The eleven constant-maturity points span one month to thirty years and carry two features that stress any interpolant: a front-end kink, the one-month yield (3.72%) sitting above the three-month (3.68%) before the curve resumes rising, and a nearly flat long end, the twenty- and thirty-year yields almost coinciding at 5.06–5.07%. We interpolate the observed yield curve directly; treating the par yields as a zero curve for the purpose of deriving instantaneous forwards is an approximation made only for the diagnostic of Section VI, and the interpolation theory is unchanged if one instead tensions a bootstrapped zero or forward curve.

III. The Natural Cubic Spline

For knots $x_0 < x_1 < \cdots < x_{n-1}$ with data y_i, the natural cubic spline is the function that is cubic on each $[x_i, x_{i+1}]$, interpolates the data, and has continuous first and second derivatives. Writing $z_i = f''(x_i)$ and $h_i = x_{i+1} - x_i$, the requirement $f^{(\mathrm{iv})} = 0$ on each interval makes f'' linear; the linear function taking the values z_i at x_i and z_{i+1} at x_{i+1} is

$$f_i''(x) = \frac{z_i}{h_i}(x_{i+1} - x) + \frac{z_{i+1}}{h_i}(x - x_i). \tag{6}$$

Integrate once, then again, carrying two constants p_i, q_i:

$$f_i'(x) = -\frac{z_i}{2h_i}(x_{i+1}-x)^2 + \frac{z_{i+1}}{2h_i}(x-x_i)^2 + p_i, \qquad f_i(x) = \frac{z_i}{6h_i}(x_{i+1}-x)^3 + \frac{z_{i+1}}{6h_i}(x-x_i)^3 + p_i(x-x_i) + q_i.$$

Imposing $f_i(x_i) = y_i$ gives $q_i = y_i - z_i h_i^2/6$, and $f_i(x_{i+1}) = y_{i+1}$ gives $p_i = (y_{i+1}-y_i)/h_i - h_i(z_{i+1}-z_i)/6$. Collecting powers of $(x - x_i)$ yields the Horner form

$$f_i(x) = y_i + (x - x_i)\big[C_i + (x - x_i)\big(B_i + (x - x_i)A_i\big)\big], \tag{7}$$

$$A_i = \frac{z_{i+1} - z_i}{6h_i}, \quad B_i = \frac{z_i}{2}, \quad C_i = \frac{y_{i+1} - y_i}{h_i} - \frac{h_i}{6}z_{i+1} - \frac{h_i}{3}z_i. \tag{8}$$

By construction f and f'' are already continuous at the interior knots (both sides share y_i and z_i). The remaining condition is continuity of f' at each x_i. From the integrated slope, the left interval $i-1$

gives $f'_{i-1}(x_i) = p_{i-1} + z_i h_{i-1}/2 = (y_i - y_{i-1})/h_{i-1} + h_{i-1}(2z_i + z_{i-1})/6$, and the right interval i gives $f'_i(x_i) = p_i - z_i h_i/2 = (y_{i+1} - y_i)/h_i - h_i(2z_i + z_{i+1})/6$. Equating $f'_{i-1}(x_i) = f'_i(x_i)$ and multiplying by 6 gives the diagonally dominant tridiagonal system, for $i = 1, \ldots, n-2$,

$$h_{i-1}z_{i-1} + 2(h_{i-1} + h_i)z_i + h_i z_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right), \tag{9}$$

with the natural boundary conditions $z_0 = z_{n-1} = 0$. The cubic spline minimises $\int_a^b \big[f''(x)\big]^2\, dx$; since the curvature of a graph is $|f''|\,[1 + (f')^2]^{-3/2}$ and $(f')^2$ is negligible for a rate curve, the cubic is effectively the minimum-curvature interpolant. Its smoothness is, however, bought at the cost of locality and — for steep or kinked data — of possible overshoot.

IV. The Tension Spline as a Tensioned Elastic Beam

A. *The physical model*

The name "spline" is borrowed from the draughtsman's tool: a thin elastic lath threaded through fixed pins. Left free, the lath assumes the minimum-bending-energy shape — the cubic spline. If the lath is additionally pulled taut by a longitudinal force T at its ends, it straightens between the pins; in the limit of infinite tension it becomes the polygonal (linear) interpolant. This is precisely the behaviour wanted of a rate curve: a tunable departure from the over-convex cubic towards the economically conservative piecewise-linear shape. We therefore model the curve as a thin elastic beam of flexural rigidity EI held under axial tension T, and we make the analogy exact rather than illustrative.

Within the small-deflection (Euler–Bernoulli) theory, the elastic energy stored by such a beam over $[a, b]$ is the sum of a flexural term and an axial (tension) term,

$$\mathcal{E}[f] = \int_a^b \Big[\tfrac{1}{2}EI\,(f'')^2 + \tfrac{1}{2}T\,(f')^2\Big]\, dx. \tag{10}$$

The first term is the bending strain energy $\int M^2/(2EI)\, dx$ with bending moment $M = EI\, f''$; the second is the work done against the axial force as the beam elongates, T times the second-order arc-length stretch $\frac{1}{2}\int (f')^2\, dx$. To see the latter, the arc length of the graph is $\int \sqrt{1 + (f')^2}\, dx = \int \big[1 + \frac{1}{2}(f')^2 + O((f')^4)\big]\, dx$, so the elongation beyond the chord is $\frac{1}{2}\int (f')^2\, dx$ to leading order, and the work done by the constant tension T against this elongation is $\frac{1}{2}T\int (f')^2\, dx$. The two energies, the moments and the deflection are shown in Figure 1.

For clarity, the symbols carry the following meaning. The function $f(x)$ is the interpolated rate curve itself: the height of the beam above the horizontal axis at abscissa x (a tenor), i.e. the rate at that tenor; its first derivative $f'(x)$ is the slope of the curve and its second derivative $f''(x)$ is the curvature. The constant E is the material's Young's modulus (stiffness of the beam material) and I is the second moment of area of its cross-section; their product EI, the *flexural rigidity*, measures how strongly the beam resists bending — a large EI yields a smooth, gently curved shape. The constant T is the axial *tension*, the longitudinal force pulling the beam taut at its ends; a large T straightens the beam between the knots. The bending moment $M(x) = EI\, f''(x)$ is the internal torque the beam carries at x. Only the dimensionless combination of the two physical constants matters for the curve shape: the *tension parameter*

$$\sigma = \sqrt{T/EI} \quad \text{(units of } 1/x\text{)},$$

which sets the single knob of the construction. When $\sigma = 0$ (no tension, $T = 0$) the functional (10) retains only the bending term and f is the natural cubic spline; as $\sigma \to \infty$ (overwhelming tension) the axial term dominates and f approaches the piecewise-linear interpolant. The limits a, b are the first and last tenors, and the $\frac{1}{2}$ factors are the usual quadratic-energy convention. Because only σ enters the Euler–Lagrange equation below, the individual values of E, I and T never need to be specified — they are absorbed into the one parameter σ.

Proposition 1. *The stationary curve of* (10) *subject to interpolation* $f(x_i) = y_i$ *satisfies, on each open interval* (x_i, x_{i+1}),

$$EI\, f^{(\mathrm{iv})} - T\, f'' = 0 \qquad \Longleftrightarrow \qquad f^{(\mathrm{iv})} - \sigma^2 f'' = 0, \qquad \sigma^2 := \frac{T}{EI}. \tag{11}$$

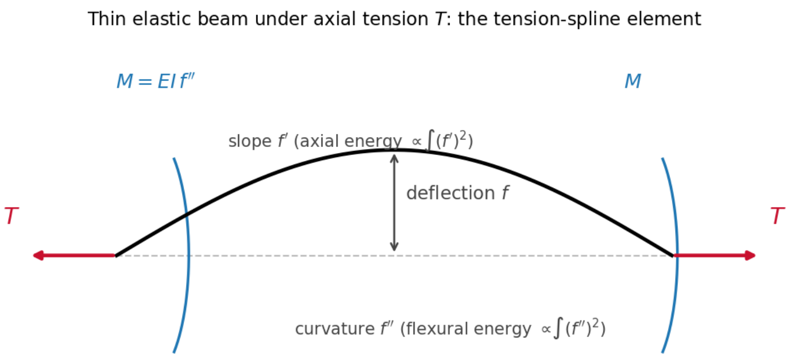

Figure 1: *The tension-spline element as a thin elastic beam under axial tension T. The bending moment $M = EIf''$ stores the flexural energy $\frac{1}{2}EI\int(f'')^2$ (the "smoothness" norm); the slope f' stores the axial energy $\frac{1}{2}T\int(f')^2$ (the "length" norm). These are the two halves of one stored elastic energy.*

Proof. Perturb $f \mapsto f + \epsilon\,\delta f$ with $\delta f(x_i) = 0$ at every knot (interpolation is fixed). Differentiating (10) and evaluating at $\epsilon = 0$,

$$\delta\mathcal{E} = \int_a^b \left[EI\,f''\,\delta f'' + T\,f'\,\delta f'\right] dx.$$

Integrate the flexural term by parts twice:

$$\int_a^b EI\,f''\,\delta f''\,dx = \left[EI\,f''\,\delta f'\right]_a^b - \int_a^b EI\,f'''\,\delta f'\,dx = \left[EI\,f''\,\delta f' - EI\,f'''\,\delta f\right]_a^b + \int_a^b EI\,f^{(\mathrm{iv})}\,\delta f\,dx,$$

and the axial term by parts once: $\int_a^b Tf'\,\delta f'\,dx = \left[Tf'\,\delta f\right]_a^b - \int_a^b Tf''\,\delta f\,dx$. The boundary terms in δf vanish because $\delta f = 0$ at the knots; the remaining boundary terms $\left[EIf''\,\delta f'\right]$ vanish under the natural free-end condition $f''(a) = f''(b) = 0$. Collecting the interior integrals,

$$\delta\mathcal{E} = \int_a^b \left[EI\,f^{(\mathrm{iv})} - T\,f''\right] \delta f\,dx.$$

Since this must vanish for every admissible δf, the fundamental lemma of the calculus of variations gives $EIf^{(\mathrm{iv})} - Tf'' = 0$ on each open interval, i.e. (11). $\square$

Equation (11) is the hyperbolic tension-spline equation of Schweikert [5] and Cline [6]; the tension parameter σ that the literature introduces by fiat is here identified as $\sigma = \sqrt{T/EI}$, the square root of the dimensionless ratio of axial to flexural stiffness. As $\sigma \to 0$ (no tension) the beam relaxes to the cubic; as $\sigma \to \infty$ it pulls taut to the linear interpolant.

B. *Construction*

The characteristic polynomial of (11) is $r^4 - \sigma^2 r^2 = r^2(r^2 - \sigma^2) = 0$, with roots $0, 0, \pm\sigma$, so the general solution on $[x_i, x_{i+1}]$ is

$$f(x) = a_i + b_i x + c_i \sinh(\sigma x) + d_i \cosh(\sigma x).$$

Differentiating twice, $f'' = \sigma^2\big(c_i \sinh\sigma x + d_i \cosh\sigma x\big)$, so f'' is a pure hyperbolic combination; writing $z_i := f''(x_i)$ and $z_{i+1} := f''(x_{i+1})$ and solving the 2×2 system for the hyperbolic part gives $f''(x) = \left[z_i \sinh(\sigma(x_{i+1} - x)) + z_{i+1} \sinh(\sigma(x - x_i))\right] / \sinh(\sigma h_i)$, which is (14) and already satisfies $f''(x_i) = z_i$, $f''(x_{i+1}) = z_{i+1}$. Integrating f'' twice and fixing the two constants of integration by the interpolation conditions $f(x_i) = y_i$, $f(x_{i+1}) = y_{i+1}$ removes the homogeneous linear part in favour of the chord, giving (12); one differentiation gives (13). Explicitly, with $h_i = x_{i+1} - x_i$,

$$f(x) = \frac{z_i \sinh\big(\sigma(x_{i+1} - x)\big) + z_{i+1} \sinh\big(\sigma(x - x_i)\big)}{\sigma^2 \sinh(\sigma h_i)} + \left(y_i - \tfrac{z_i}{\sigma^2}\right)\frac{x_{i+1} - x}{h_i} + \left(y_{i+1} - \tfrac{z_{i+1}}{\sigma^2}\right)\frac{x - x_i}{h_i}, \tag{12}$$

$$f'(x) = \frac{-z_i \cosh\big(\sigma(x_{i+1} - x)\big) + z_{i+1} \cosh\big(\sigma(x - x_i)\big)}{\sigma \sinh(\sigma h_i)} + \frac{(y_{i+1} - y_i) - (z_{i+1} - z_i)/\sigma^2}{h_i}, \tag{13}$$

$$f''(x) = \frac{z_i \sinh\big(\sigma(x_{i+1} - x)\big) + z_{i+1} \sinh\big(\sigma(x - x_i)\big)}{\sinh(\sigma h_i)}. \tag{14}$$

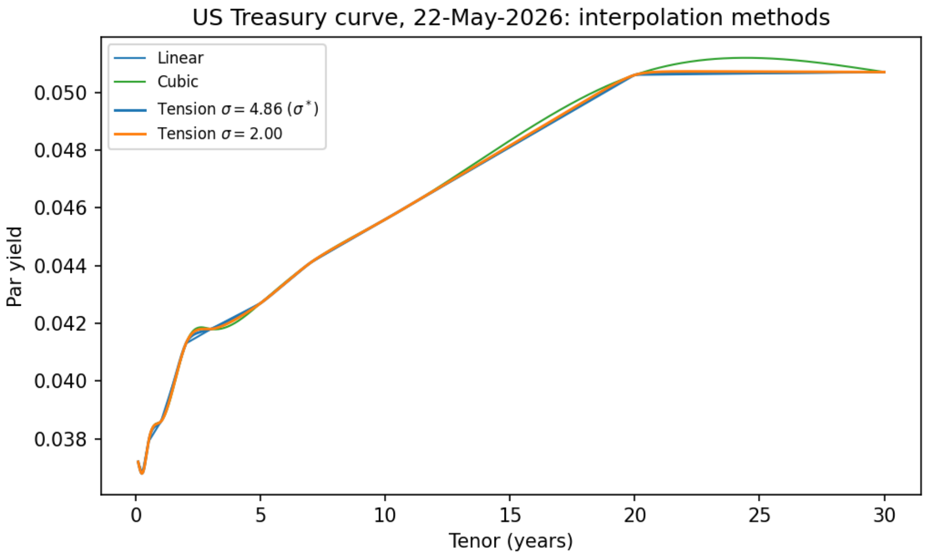

Figure 2: *Linear, cubic, and tension-spline interpolation of the U.S. Treasury par-yield curve of Table 1. The tension curve at the equipartition optimum $\sigma^\star = 4.86$ (Section V) removes the cubic's spurious long-end hump while retaining a smooth belly.*

The construction makes f and f'' continuous at the interior knots automatically (both sides share y_i and z_i); the remaining $n-2$ unknowns $z_1, \dots, z_{n-2}$ are fixed by requiring continuity of f'. Evaluating (13) at x_i^- (from interval $i-1$) and x_i^+ (from interval i) and equating yields, after collecting the coefficients of z_{i-1}, z_i, z_{i+1}, the diagonally dominant tridiagonal system, for $i = 1, \dots, n-2$,

$$\alpha_{i-1} z_{i-1} + (\beta_{i-1} + \beta_i) z_i + \alpha_i z_{i+1} = \gamma_i - \gamma_{i-1}, \tag{15}$$

$$\alpha_i = \frac{1}{h_i} - \frac{\sigma}{\sinh(\sigma h_i)}, \qquad \beta_i = \frac{\sigma \cosh(\sigma h_i)}{\sinh(\sigma h_i)} - \frac{1}{h_i}, \qquad \gamma_i = \frac{\sigma^2 (y_{i+1} - y_i)}{h_i}, \tag{16}$$

with $z_0 = z_{n-1} = 0$. As $\sigma \to 0$, Taylor expansion gives $\alpha_i \to -h_i/6$, $\beta_i \to h_i/3$ and $\gamma_i \to (y_{i+1} - y_i)/h_i$ (times σ^2, which cancels), so (15) reduces to the natural-cubic system of Section III; numerically at $\sigma = 10^{-3}$ the two interpolants of Table 1 agree to 3×10^{-9}. The implementation differs from the cubic only in the hyperbolic evaluations; for large σh_i the formulae are reorganised into the stable $e^{-\sigma h}$ form of Appendix C to avoid overflow. Figure 2 shows the linear, cubic, and two tension interpolants of the yield curve; smaller σ lies near the cubic and larger σ approaches the linear shape. Already in yield space the cubic overshoots: it lifts the flat 20–30-year segment into a spurious hump of roughly five basis points peaking near twenty-four years, a feature absent from the data, whereas the tension curve hugs the flat segment.

V. Energy Decomposition and Tension Selection

A. *The two energies are the two norms*

The literature scores a candidate curve by a smoothness norm $\int (f'')^2$ and, in tension schemes, by a first-derivative norm weighted by σ^2. Proposition 1 identifies these immediately. Define, per unit flexural rigidity, the *flexural* and *axial* energies of the tensioned beam,

$$E_{\text{bend}}(\sigma) = \tfrac{1}{2} \int_0^T \left(f_\sigma''\right)^2 dt, \qquad E_{\text{tens}}(\sigma) = \tfrac{1}{2}\, \sigma^2 \int_0^T \left(f_\sigma'\right)^2 dt. \tag{17}$$

The smoothness norm *is* $2E_{\text{bend}}$ and the σ^2-weighted length norm *is* $2E_{\text{tens}}$; their sum is the total stored elastic energy $\mathcal{E}[f_\sigma]$ of (10). They are therefore not independent scores to be balanced against each other by an external rule, but the two complementary components of one energy. This observation also dissolves a circularity in earlier "beam-energy" arguments that compared a curvature norm against a separately constructed point-load deflection energy: by Clapeyron's theorem the work done by a load equals the strain energy it stores, so the two were estimates of the same flexural quantity, and the point-load route

— which requires inverting for the location and magnitude of a fictitious load — is both redundant and ill-posed for the S-shaped segments a spline can produce. We discard it in favour of (17).

Both integrals in (17) are elementary. Using (14) and the identities $\int_0^h \sinh^2(\sigma u)\,du = \frac{\sinh(2\sigma h)}{4\sigma} - \frac{h}{2}$ and $\int_0^h \sinh(\sigma(h-u))\sinh(\sigma u)\,du = \frac{1}{2}\big(h\cosh(\sigma h) - \sigma^{-1}\sinh(\sigma h)\big)$, the per-segment flexural energy is

$$E_{\text{bend}}^{(i)} = \frac{1}{2\sinh^2(\sigma h_i)}\left[(z_i^2 + z_{i+1}^2)\left(\frac{\sinh(2\sigma h_i)}{4\sigma} - \frac{h_i}{2}\right) + z_i z_{i+1}\left(h_i\cosh(\sigma h_i) - \frac{\sinh(\sigma h_i)}{\sigma}\right)\right]. \tag{18}$$

The axial energy admits the companion closed form (34) of Appendix B. Both expressions were verified against high-order quadrature to a relative error of 3×10^{-7}. The total energies are $E_{\text{bend}} = \sum_i E_{\text{bend}}^{(i)}$ and likewise for E_{tens}.

B. The selection criterion

Every value of σ already minimises its own functional (10); the optimal tension is therefore a question *outside* the variational principle, and a selection rule must supply an external preference. We make that preference dimensionless and explicit through the *bending-energy fraction*

$$\varphi(\sigma) = \frac{E_{\text{bend}}(\sigma)}{E_{\text{bend}}(\sigma) + E_{\text{tens}}(\sigma)} \in (0,1). \tag{19}$$

At $\sigma \to 0$ the axial energy vanishes and $\varphi \to 1$: the stored energy is pure curvature, the cubic limit. At $\sigma \to \infty$ the interior curvature vanishes and $\varphi \to 0$: the energy is pure tension, the linear limit. The following proposition makes the well-posedness of the selection precise.

Proposition 2 (Properties of φ)**.** *Write $t = \sigma^2$, $B(\sigma) = \int (f_\sigma'')^2$, $A(\sigma) = \int (f_\sigma')^2$, so $E_{\text{bend}} = \frac{1}{2}B$, $E_{\text{tens}} = \frac{1}{2}tA$ and $\varphi(\sigma) = B/(B+tA)$. Then*

(i) φ is continuous on $(0,\infty)$, with $\lim_{\sigma\to0^+}\varphi = 1$ and $\lim_{\sigma\to\infty}\varphi = 0$;

(ii) consequently, for every target $\varphi^\star \in (0,1)$ there exists $\sigma^\star > 0$ with $\varphi(\sigma^\star) = \varphi^\star$;

(iii) the minimum stored energy $g(t) = \min_f\{B[f]+tA[f]\} = B(\sigma)+tA(\sigma)$ is concave in t, and $A(\sigma) = g'(t)$ is non-increasing in σ;

(iv) φ is strictly decreasing wherever $A\,B + t\,g''(t)\,g(t) > 0$; this holds throughout the range of interest, so $\sigma^\star$ is unique.

Proof. (i) The map $\sigma \mapsto f_\sigma$ is continuous (the tridiagonal system (15) has coefficients analytic in σ and is uniformly invertible), so B, A are continuous and φ is continuous where $B + tA > 0$, i.e. for all $\sigma > 0$. As $\sigma \to 0^+$, f_σ tends to the natural cubic with $0 < B \to B_{\text{cub}} < \infty$ and $A \to A_{\text{cub}} < \infty$, so $tA \to 0$ and $\varphi \to 1$. As $\sigma \to \infty$, f_σ tends to the piecewise-linear interpolant: $A \to \int (f_{\text{lin}}')^2 > 0$ is bounded below, while the curvature concentrates in boundary layers of width $O(1/\sigma)$ about the knots where $f'' = O(\sigma)$, so $B = O(\sigma)$. Hence $tA/B = O(\sigma^2)/O(\sigma) \to \infty$ and $\varphi = 1/(1+tA/B) \to 0$. *(ii)* Immediate from (i) and the intermediate value theorem. *(iii)* For each fixed admissible f, $t \mapsto B[f]+tA[f]$ is affine with slope $A[f] \ge 0$; g is the pointwise minimum of this family and is therefore concave. By the envelope theorem $g'(t) = A[f_t] = A(\sigma)$, and concavity gives $g'' = A' \le 0$, so A is non-increasing. *(iv)* Let $R = tA/B = E_{\text{tens}}/E_{\text{bend}}$, so $\varphi = 1/(1+R)$ and φ decreases iff R increases. Writing $A = g'$ and $B = g - tg'$,

$$R'(t) = \frac{d}{dt}\frac{tg'}{g-tg'} = \frac{g'(g-tg') + t\,g''g}{(g-tg')^2} = \frac{A\,B + t\,g''g}{B^2},$$

which is positive precisely under the stated condition. Since $g'' \le 0$ the condition can fail only if the trade-off curve is strongly concave; on the curve of Table 1 we verify $d\varphi/d\sigma < 0$ at every point of a dense grid, and (i) guarantees the endpoints, so $\sigma^\star$ is unique. □

We state plainly what is and is not proved: parts (i)–(iii) hold for any non-degenerate data; the strict global monotonicity (iv) is reduced to an explicit sign condition that we verify numerically rather than assert as an unconditional theorem, since for adversarial knot configurations the trade-off curve g could in principle be concave enough to violate it. Existence of $\sigma^\star$ is unconditional; uniqueness is guaranteed under (iv).

Definition 1 (Energy-partition tension)**.** *Given a target fraction $\varphi^\star \in (0,1)$, the optimal tension $\sigma^\star$ is the root of $\varphi(\sigma) = \varphi^\star$. The neutral choice $\varphi^\star = \frac{1}{2}$ corresponds to* equipartition *of elastic energy between flexure and tension.*

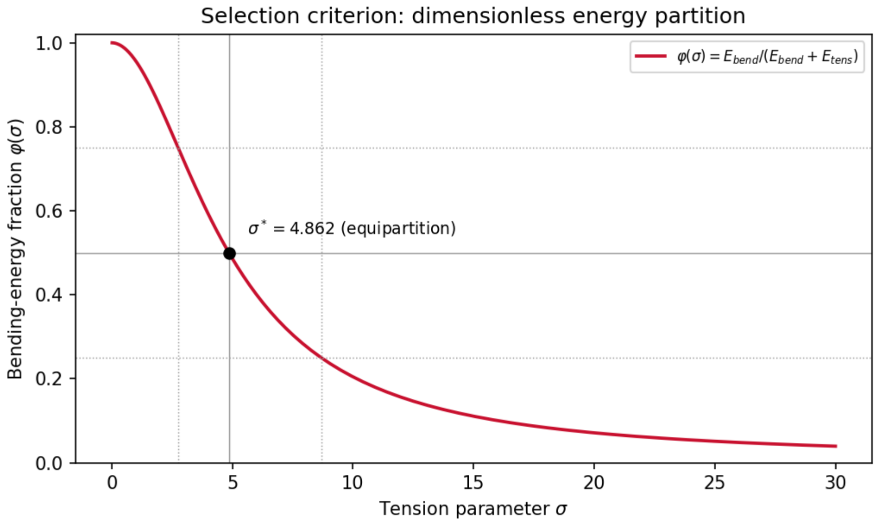

Figure 3: *The bending-energy fraction $\varphi(\sigma)$ for the curve of Table 1. It descends smoothly and monotonically from one (cubic) to zero (linear); equipartition ($\varphi = \frac{1}{2}$) is attained at $\sigma^\star = 4.86$. The dotted lines mark $\varphi^\star = 0.75$ ($\sigma = 2.76$) and $\varphi^\star = 0.25$ ($\sigma = 8.69$). Contrast the monotone single crossing here with the spiky, range-dependent energy curves produced by the point-load construction.*

Two further properties recommend (19) over earlier max-normalised rules. First, it is *range-independent*: φ is an intrinsic property of each σ and does not depend on the window of σ over which a normalisation is taken. Second, it is *scale-invariant.*

Proposition 3 (Scale invariance)**.** *$\varphi(\sigma)$ is invariant under rescaling of the rate axis, and under joint rescaling of the time axis with σ carried in inverse-time units.*

Proof. Rate axis. Replace $y_i \mapsto \alpha y_i$. The tridiagonal right-hand side $\gamma_i - \gamma_{i-1}$ in (15) scales by α while the matrix is unchanged, so $z_i \mapsto \alpha z_i$ and hence $f \mapsto \alpha f$, $f' \mapsto \alpha f'$, $f'' \mapsto \alpha f''$. Then $B = \int (f'')^2 \mapsto \alpha^2 B$ and $A = \int (f')^2 \mapsto \alpha^2 A$, so both energies scale by α^2 and $\varphi = B/(B + \sigma^2 A)$ is unchanged. This is the basis-points-versus-decimals invariance. *Time axis.* Replace $x \mapsto \beta x$ (so $h_i \mapsto \beta h_i$) and $\sigma \mapsto \sigma/\beta$, which keeps every product σh_i fixed; the shape of f in the (x, y) plane is therefore unchanged, only horizontally stretched. Under $x \mapsto \beta x$ a derivative carries a factor β^{-1}: $f' \mapsto \beta^{-1} f'$, $f'' \mapsto \beta^{-2} f''$, while $dx \mapsto \beta\, dx$. Thus $B = \int (f'')^2 dx \mapsto \beta^{-4} \cdot \beta\, B = \beta^{-3} B$, and with $\sigma^2 \mapsto \beta^{-2}\sigma^2$, $\sigma^2 A = \sigma^2 \int (f')^2 dx \mapsto \beta^{-2} \cdot \beta^{-2} \cdot \beta\, \sigma^2 A = \beta^{-3}\sigma^2 A$. Both numerator and denominator scale by β^{-3}, so φ is unchanged. This is the days-versus-years invariance. $\square$

This removes the unit dependence that afflicts an unnormalised energy criterion, under which a choice of basis points versus decimals, or days versus years, would silently move the optimum. The target $\varphi^\star$ is the single risk–hedging dial called for in the introduction: $\varphi^\star \to 1$ requests a smoother, cubic-like curve; $\varphi^\star \to 0$ buys locality and forward stability at the cost of curvature. The procedure is summarised as follows.

1. Choose a target fraction $\varphi^\star$ (default $\frac{1}{2}$).
2. For trial σ, solve (15) for $\{z_i\}$ and evaluate $E_{\text{bend}}(\sigma)$ and $E_{\text{tens}}(\sigma)$ in closed form via (18) and (34).
3. Form $\varphi(\sigma)$ from (19).
4. Solve $\varphi(\sigma^\star) = \varphi^\star$ by bisection (one-dimensional, monotone).
5. Return $\sigma^\star$ and the curve $f_{\sigma^\star}$.

C. Results

For the Treasury curve of Table 1 the criterion gives $\sigma^\star = 2.76$ at $\varphi^\star = 0.75$, $\sigma^\star = 4.86$ at the equipartition default, and $\sigma^\star = 8.69$ at $\varphi^\star = 0.25$. Figure 4 plots the two component energies. The flexural energy is nearly flat in σ — the data's curvature is concentrated in the densely knotted short end, which does not straighten until very high tension — while the axial energy climbs from zero and crosses E_{bend} exactly at the equipartition point. The crossover is therefore driven by the growth of the tension energy against a slowly varying bending energy, and the optimum is well separated from both spline limits.

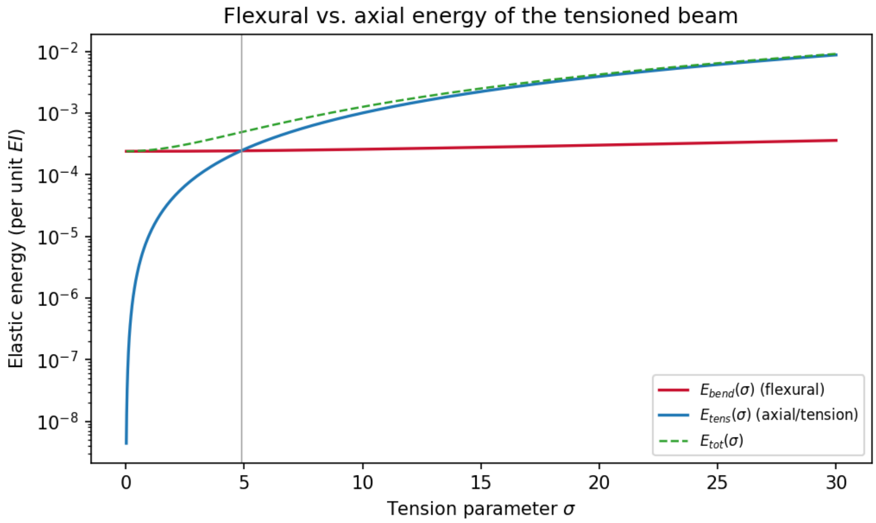

Figure 4: *Flexural energy E_{bend} and axial energy E_{tens} (log scale, per unit EI) versus σ. The crossing $E_{\text{bend}} = E_{\text{tens}}$ is the equipartition optimum $\sigma^\star = 4.86$.*

VI. Validation

An interpolating spline reproduces every input instrument exactly, so the in-sample repricing error is identically zero and cannot discriminate between values of σ. The discriminating tests are the no-arbitrage and hedging properties that motivated the construction.

Forward positivity and overshoot. A necessary no-arbitrage condition is that the implied instantaneous forward $f(t) = y(t) + t\,y'(t)$ be non-negative. On the curve of Table 1 both the cubic and the tension interpolant keep forwards comfortably positive (minimum 3.64%), as one expects of a steep upward curve; this data set does not trigger the negative-forward pathology, and we do not manufacture one. The discriminating feature lies elsewhere, in the long end (Figure 5). The flat 20–30-year par segment implies a near-flat forward of about 5.07%, which the tension curve reproduces. The cubic instead fabricates a forward hump rising to almost 6% near eighteen years and then declining to 4.7% at thirty years — a shape with no support in the data, the same overshoot that is visible directly in the yield curve of Figure 2. The price of the tension curve's fidelity is a kink in the forward at twenty years, where the steep 10–20-year segment meets the flat 20–30-year one; this is the smoothness-versus-fidelity trade-off that $\varphi^\star$ governs, and it is also a reminder, in the spirit of Hagan and West [8], that when forward smoothness is itself the objective one should tension the forward curve rather than the yield curve — the area-constrained construction we develop in Section VII.

Hedge locality. The economically decisive property is locality: how far along the curve a perturbation of a single input instrument propagates. We perturb the forward at node t_k by ε, rebuild the curve at fixed σ, and measure the influence $\partial f(t)/\partial r_k$ and the *locality radius*

$$\rho_k = \frac{\int |t - t_k|\,\big|\partial f(t)/\partial r_k\big|\,dt}{\int \big|\partial f(t)/\partial r_k\big|\,dt}, \tag{20}$$

the mean distance of the response from the perturbed node; smaller ρ_k means a more local, more easily hedged curve. Figure 6 bumps the benchmark 5-year node. The cubic response stays between ten and forty percent of its peak across the *entire* thirty-year curve — secondary lobes near thirteen and twenty-four years — giving $\rho_5 = 6.7$ years, whereas the tension spline at $\sigma^\star$ decays roughly exponentially in $\sigma|t - t_k|$ and is effectively extinguished beyond ten years, confining the response to $\rho_5 = 0.8$ year, an eight-fold improvement. The gain is consistent across the belly of the curve: the locality ratio between cubic and tension ranges from about seven at the front to nine at two years, narrowing only at the sparsely knotted 20–30-year end, where the ten-year gaps between nodes limit the locality attainable by any interpolant. The energy-partition tension thus delivers precisely the local hedges that the introduction posed as the goal, and does so through a single, interpretable parameter.

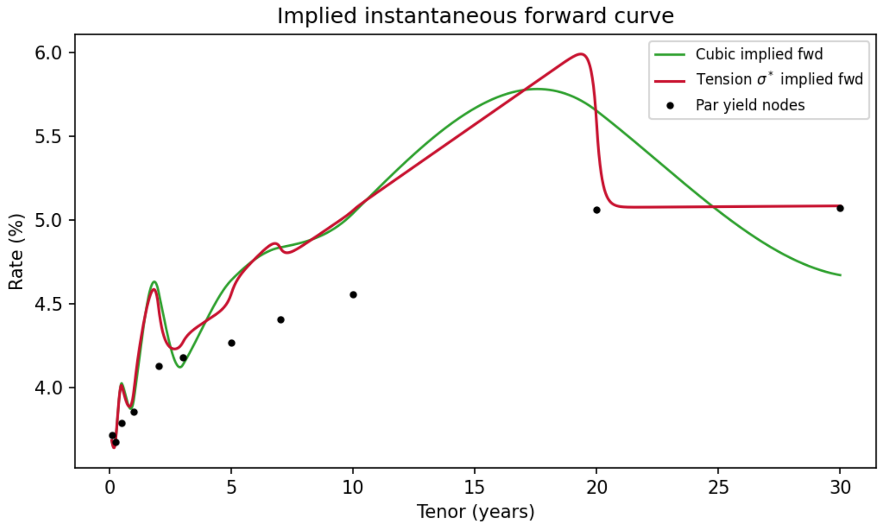

Figure 5: *Implied instantaneous forwards. The cubic manufactures a forward hump to nearly* 6% *and a declining long end unsupported by the flat* 20–30*-year par data; the tension curve at* $\sigma^\star$ *tracks the data-implied flat forward, at the cost of a kink at twenty years.*

The exponential decay has a clean physical reading. The influence of a perturbation is governed by the Green's function of the spline operator $\mathcal{L} = d^4/dx^4 - \sigma^2 d^2/dx^2$, i.e. the solution of $\mathcal{L}\, G(x, x_k) = \delta(x - x_k)$. Away from the source this satisfies the homogeneous equation $G^{(\mathrm{iv})} - \sigma^2 G'' = 0$, whose decaying mode is $e^{-\sigma|x-x_k|}$. Thus σ is an inverse screening length: a rate shock at t_k is an evanescent disturbance that dies out over a distance $\sim 1/\sigma$, which is exactly the hedge-locality radius ρ_k. In the untensioned ($\sigma \to 0$) cubic limit the operator degenerates to d^4/dx^4, whose Green's function decays only algebraically and oscillates — the long-range "ringing" of Figure 6 — so a shock propagates across the whole curve like an undamped wave. Tension plays the role of a restoring force that damps this propagation; raising σ shortens the screening length and localises the hedge.

VII. Tensioning the Instantaneous Forward Curve

The validation of Section VI exposed a structural point. Smoothness is a property of the interpolated variable and is degraded by differentiation: if the zero curve $r(t)$ is the C^2 tension spline, the instantaneous forward $f(t) = r(t) + t\, r'(t)$ is only C^1, and as σ grows and r approaches piecewise-linear, f develops kinks at the knots. The three natural interpolation variables, linked by $I(t) := -\ln P(t) = r(t)\, t = \int_0^t f$, behave differently under tension in the high-σ limit: tensioning r gives a kinked piecewise-linear forward; tensioning I (the log-discount) gives $f = I'$, a piecewise-*constant* forward (the familiar local "constant-forward" rule); and tensioning f itself keeps the forward a continuous C^2 spline for every σ. Hence, *if forward smoothness is the objective, the forward curve is the object to tension.*

The forward is not observed pointwise; the market pins the discount factors, equivalently the integrals of f over each interval. Tensioning the forward is therefore an *area-constrained* variational problem: minimise the forward's own tension energy subject to reproducing every discount factor,

$$\min_f \int_0^T \Big[\tfrac{1}{2}\big(f''\big)^2 + \tfrac{1}{2}\sigma^2\big(f'\big)^2\Big]\, dt \quad \text{s.t.} \quad \int_{t_{i-1}}^{t_i} f(u)\, du = I(t_i) - I(t_{i-1}) =: a_i, \quad i = 1, \dots, m. \tag{21}$$

With a multiplier λ_i for each area constraint, form the Lagrangian

$$\mathcal{L}[f, \lambda] = \int_0^T \Big[\tfrac{1}{2}(f'')^2 + \tfrac{1}{2}\sigma^2 (f')^2\Big] dt - \sum_{i=1}^m \lambda_i \Big(\int_{t_{i-1}}^{t_i} f\, du - a_i \Big).$$

The constraint integral contributes $-\lambda_i$ to the variation on (t_{i-1}, t_i), while the energy term varies exactly as in Proposition 1 (integrate $f''\delta f''$ by parts twice and $\sigma^2 f'\delta f'$ once). Setting $\delta\mathcal{L} = 0$ gives, on each interval, the Euler–Lagrange equation with a piecewise-constant forcing,

$$f^{(\mathrm{iv})} - \sigma^2 f'' = \lambda_i \quad \text{on } (t_{i-1}, t_i), \tag{22}$$

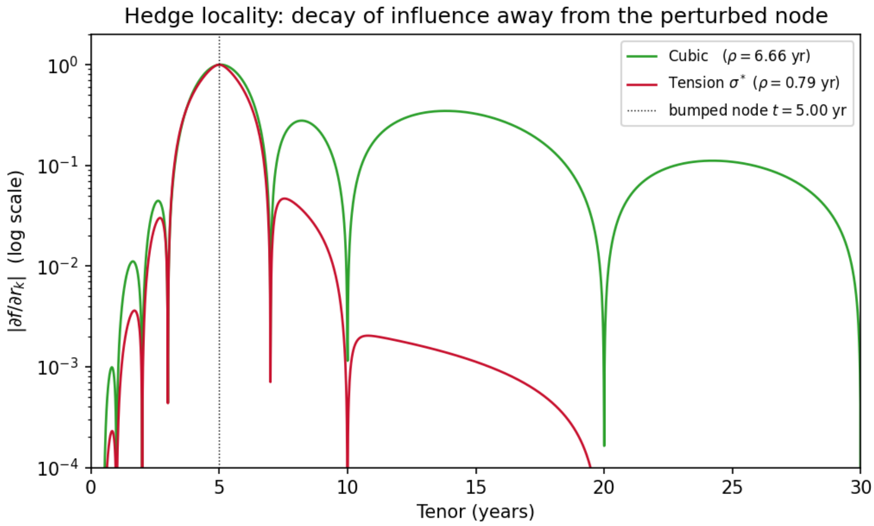

Figure 6: *Decay of the single-node influence $|\partial f/\partial r_k|$ (log scale) for a bump at the benchmark 5-year node. The cubic (locality radius $\rho = 6.7$ yr) propagates the perturbation across the whole curve; the tension spline at $\sigma^\star$ ($\rho = 0.8$ yr) localises it within a few years.*

whose solution is the homogeneous tension spline plus the particular solution $f_p = -\lambda_i t^2/(2\sigma^2)$ (since $f_p^{(\mathrm{iv})} = 0$ and $-\sigma^2 f_p'' = \lambda_i$). The m multipliers are fixed by the m area constraints, together with continuity of f, f', f'' across knots and the natural end conditions $f'' = f''' = 0$. In practice the problem is discretised on a fine grid $\{t_k\}$: with D_1, D_2 the first- and second-difference operators, the energies become the quadratic forms $\int (f')^2 \approx f^\top D_1^\top D_1 f\,\Delta t =: f^\top K_t f$ and $\int (f'')^2 \approx f^\top K_b f$, and the area constraints become $Cf = a$ with C_{ik} the length of grid cell k inside interval i. Minimising $\frac{1}{2} f^\top (K_b + \sigma^2 K_t) f$ subject to $Cf = a$ has the first-order (KKT) conditions $(K_b + \sigma^2 K_t) f + C^\top \lambda = 0$ and $Cf = a$, i.e. the single sparse symmetric system

$$\begin{pmatrix} K_b + \sigma^2 K_t & C^\top \\ C & 0 \end{pmatrix} \begin{pmatrix} f \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ a \end{pmatrix}. \tag{23}$$

The energy-partition criterion carries over verbatim, with $\varphi(\sigma)$ now formed from the forward's flexural energy $\int (f'')^2$ and axial energy $\sigma^2 \int (f')^2$; tensioning the forward thus balances forward *curvature* against forward *slope* directly.

On the curve of Table 1 this construction reproduces every discount factor to 2×10^{-10} (machine precision; no-arbitrage is exact), keeps the forward positive, and returns a forward-space equipartition tension $\sigma^\star = 6.78$. Figure 7 compares, at a common tension, the forward implied by tensioning the yield curve with the forward obtained by tensioning the forward directly. The two share the front-end wiggle, which is genuine signal forced by the kinked short-end par data, but in the long end, where the data are smooth, the difference is stark: the yield-tensioned forward cliffs at twenty years, whereas the forward-tensioned curve is smooth there. Quantitatively, over $t > 8$ years the forward-tensioned curve has 1,285 times lower roughness $\int (f'')^2$ and a peak curvature at twenty years smaller by a factor of 338, at no cost in repricing. When smooth forwards matter — for forward-rate-dependent payoffs, convexity adjustments, or volatility-surface inputs — this is the construction to use; the energy-partition criterion selects its tension exactly as before.

VIII. Non-Uniform Tension Calibration

A single global tension treats every segment alike, yet the segments are not: on the curve of Table 1 the interval widths h_i span more than two orders of magnitude, from one month to ten years. The dimensionless group that fixes a segment's shape is the product σh_i, the argument of the hyperbolic functions in (12); under a uniform σ this product ranges from 0.8 on the shortest segment to 48.6 on the 20–30-year segment. The short end is therefore barely tensioned while the long end is pulled essentially straight.

The imbalance shows cleanly in the per-segment energy fraction $\varphi_i = E_{\mathrm{bend}}^{(i)}/(E_{\mathrm{bend}}^{(i)} + E_{\mathrm{tens}}^{(i)})$, the local analogue of (19). At the global optimum $\sigma^\star = 4.86$ the fractions run from 0.86 on the shortest segment down to below 0.01 on the segments between three and twenty years (Figure 8, green): the belly and long

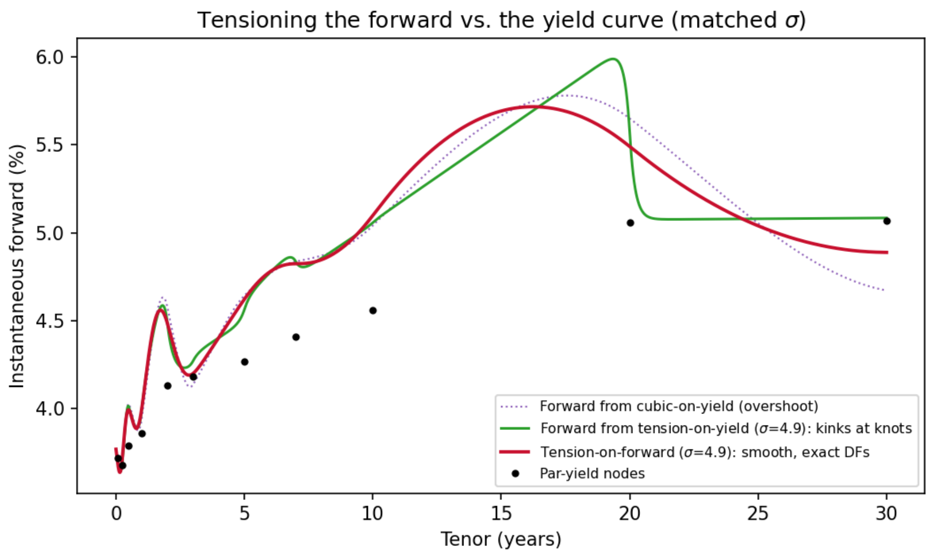

Figure 7: *Instantaneous forward at a common tension $\sigma = 4.9$. Tensioning the yield curve (green) produces a forward that cliffs at twenty years; tensioning the forward directly (red) keeps it smooth while reproducing all discount factors exactly. The cubic-on-yield forward (dotted) overshoots. All three agree on the data-driven front-end wiggle.*

end carry almost no flexural energy — they have been flattened to near-linearity — while the short end stays curved. Equipartition holds only on average, not segment by segment.

The remedy is to let the tension vary by segment. We assign a tension σ_i to each interval (the non-uniform spline of Appendix A) and calibrate the vector $\{\sigma_i\}$ so that every segment individually attains the target fraction,

$$\varphi_i(\sigma_i) = \varphi^\star \qquad \text{for all } i. \tag{24}$$

Because the per-segment energies (18)–(34) and the knot second derivatives $\{z_i\}$ are coupled through the tridiagonal system, (24) is solved by a short fixed-point iteration: holding $\{z_i\}$ fixed, each one-dimensional equation $\varphi_i(\sigma_i) = \varphi^\star$ is solved by bisection (each φ_i is monotone in its σ_i), the non-uniform tridiagonal is then re-solved for $\{z_i\}$, and the step is repeated. On the data of Table 1 the iteration converges in about forty steps and drives every φ_i to $\frac{1}{2}$ to machine precision (Figure 8, red).

The calibrated tensions (Figure 9) scale roughly as $\sigma_i \sim 1/h_i$, from $\sigma \approx 12$ on the shortest segment to $\sigma \approx 0.14$ in the belly, so that the product $\sigma_i h_i$ is compressed from the uniform 60 : 1 spread into a narrow band of order unity across the active part of the curve. Calibration thus discovers that the natural tension scale is local: each segment is tensioned in proportion to its own width and curvature rather than by a single number imposed on the whole curve. The resulting curve (Figure 10) restores the gentle curvature to the five-to-twenty-year belly that the uniform spline had straightened, while still suppressing the cubic's long-end hump. It remains C^2 and reprices every node exactly, since calibration changes only the tension profile and not the interpolation conditions.

IX. Synthesis: Non-Uniform Tension on the Forward Curve

The two extensions just developed are orthogonal and combine. Section VII makes the instantaneous forward the tensioned object, under area constraints that reproduce the discount factors; Section VIII lets the tension vary by segment to balance energy locally. Applying the second to the first gives a *non-uniform tension on the forward curve*: the forward solves the area-constrained problem (21) with a piecewise tension $\sigma(t) = \sigma_i$ on each interval, and the $\{\sigma_i\}$ are calibrated to per-segment equipartition of the forward's own energy.

The forward formulation makes the calibration markedly simpler than its yield-curve counterpart. In the discretised problem the forward f is the direct unknown, so for a *fixed* f the per-segment flexural and (unweighted) axial energies $B_i = \frac{1}{2}\int_{t_{i-1}}^{t_i} (f'')^2$ and $A_i = \frac{1}{2}\int_{t_{i-1}}^{t_i} (f')^2$ are quadratic forms in f that do not involve σ at all; the tension enters the segment's axial energy only through the multiplicative factor σ_i^2, as

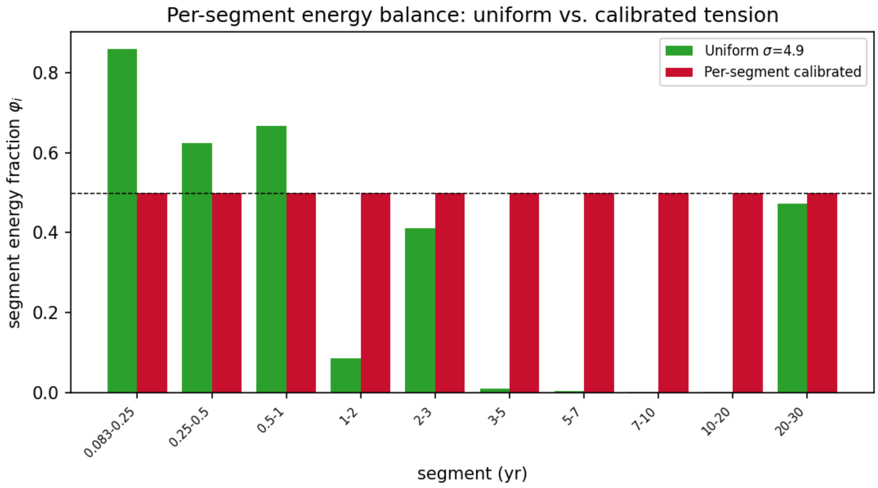

Figure 8: *Per-segment energy fraction φ_i. A uniform tension (green) leaves the fractions scattered from* 0.86 *to below* 0.01*; per-segment calibration (red) equalises them to* $\frac{1}{2}$ *on every segment.*

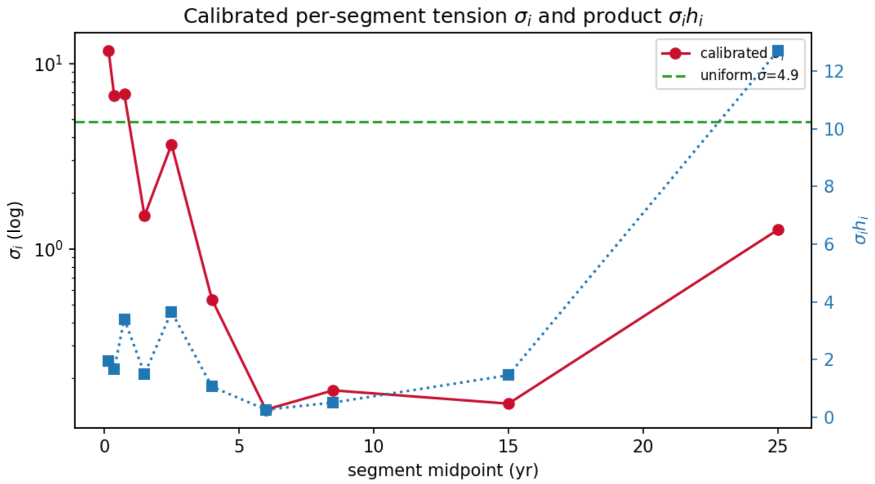

Figure 9: *Calibrated per-segment tension σ_i (red, log scale) against the uniform optimum (green dashed), and the dimensionless product $\sigma_i h_i$ (blue). Calibration raises the short-end tension and lowers the long-end tension, bringing $\sigma_i h_i$ into a narrow band.*

$E^{(i)}_{\text{tens}} = \sigma_i^2 A_i$, while $E^{(i)}_{\text{bend}} = B_i$. Write the per-segment fraction explicitly,

$$\varphi_i(\sigma_i) = \frac{E^{(i)}_{\text{bend}}}{E^{(i)}_{\text{bend}} + E^{(i)}_{\text{tens}}} = \frac{B_i}{B_i + \sigma_i^2 A_i},$$

and set it to the target $\frac{1}{2}$. Then $B_i + \sigma_i^2 A_i = 2B_i$, i.e. $\sigma_i^2 A_i = B_i$, so

$$\sigma_i^2 A_i = B_i \qquad \Longrightarrow \qquad \sigma_i = \sqrt{B_i/A_i}, \tag{25}$$

a closed-form update with no bisection required (contrast the yield-curve case of Section VIII, where the z_i depend on σ_i through the hyperbolic energies (18)–(34) and a one-dimensional root-find is needed per segment). The construction is a short fixed point: solve the KKT system for f at the current $\{\sigma_i\}$, recompute $\{B_i, A_i\}$, update each σ_i by (25), and repeat. On the curve of Table 1 it converges in about forty iterations, reproduces every discount factor to 4×10^{-11}, keeps the forward positive, and drives all eleven per-segment fractions to $\frac{1}{2}$, from the scatter of 0.001 to 0.88 left by a uniform forward tension (Figure 11, right). The calibrated tensions are large at the curved front end ($\sigma \approx 20$ on the 0.08–0.25-year segment) and small on the flat long end ($\sigma \approx 0.09$ on 20–30 years).

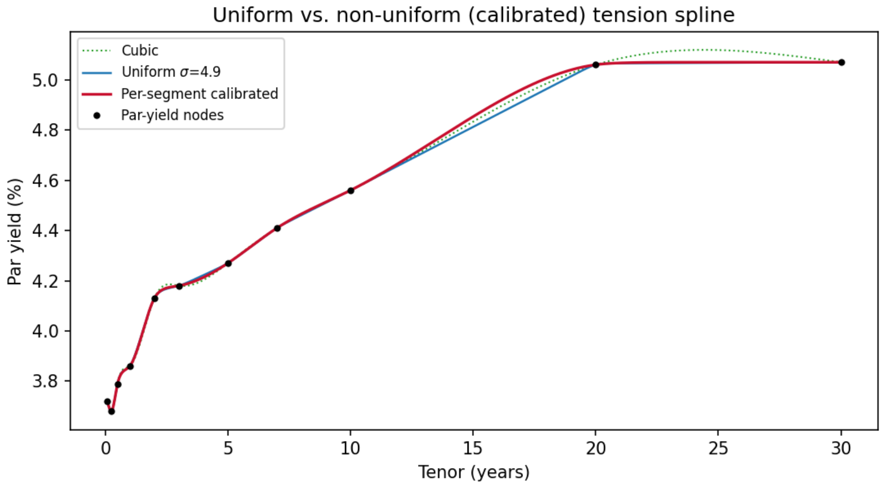

Figure 10: *Uniform versus per-segment-calibrated tension spline. The calibrated curve recovers curvature in the belly that the uniform spline flattens, while both avoid the cubic's long-end hump.*

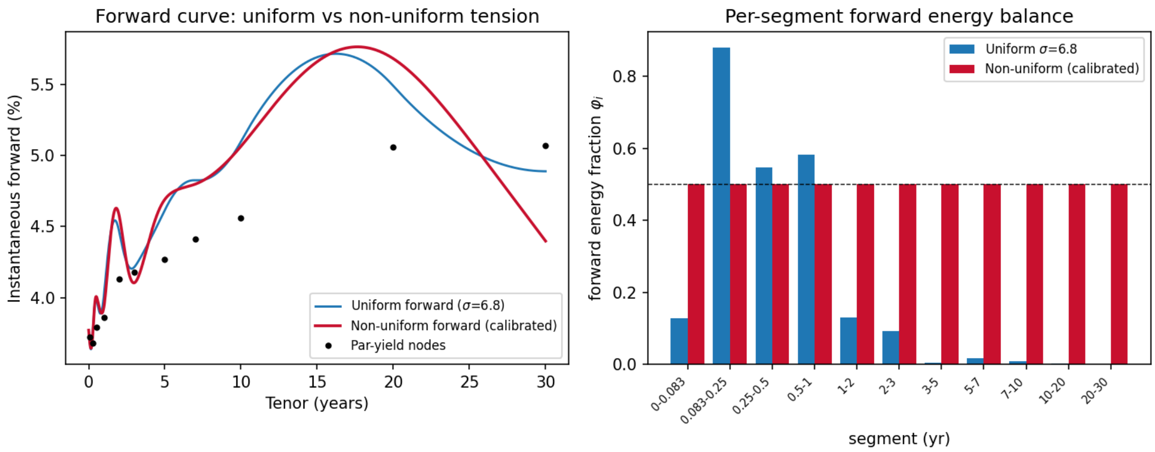

Figure 11: *Non-uniform tension on the forward curve.* Left: *the uniform and per-segment-calibrated forwards, both smooth and both reproducing all discount factors exactly.* Right: *per-segment forward energy fractions; calibration replaces the uniform tension's scatter with exact equipartition on every segment.*

One feature of this combination deserves an honest statement. Per-segment equipartition is an energy-*balance* rule, not a forward-*smoothness* rule. On a near-flat long-end segment both B_i and A_i are small, and (25) returns a *low* tension, which lets the forward curve freely over that interval rather than holding it taut (Figure 11, left, beyond twenty years). This is the correct behaviour for energy balance, but a desk that prizes a stable long-end forward would impose a lower $\varphi^\star$ there — more tension, a straighter forward — exactly the risk–preference dial of Section V, now exercised per segment. The unified construction thus delivers a C^2, exactly-repricing, energy-balanced forward curve in which a single interpretable parameter, applied globally or segment by segment, governs the whole trade-off between smoothness, locality, and long-end stability.

X. Multi-Curve Construction: SOFR Discounting and a Tensioned Projection Curve

Modern fixed-income desks operate in a multi-curve world: cashflows are discounted on an overnight-indexed (OIS / SOFR) curve, while forward rates are read off one or more separate projection curves that may carry a tenor or credit basis [2, 3, 4]. The interpolation method of this paper is deliberately agnostic to that architecture — it acts *downstream* of the bootstrap — and we now show it in place, with OIS discounting held exact and the energy-partition tension applied only to the projection curve.

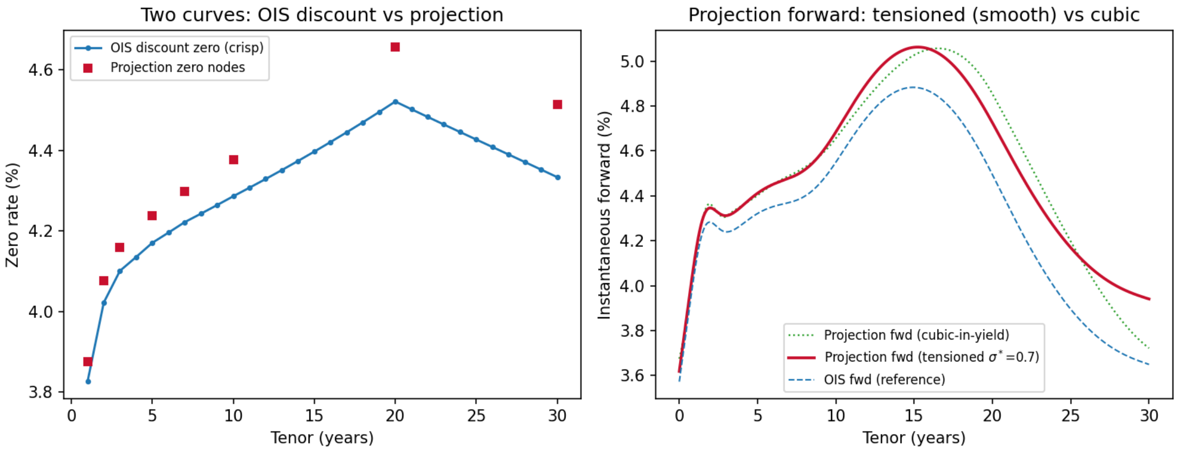

Figure 12: *Multi-curve SOFR/OIS construction.* Left: *the bootstrapped OIS discount zero curve (crisp, exact at every node) and the projection-curve nodes offset by the tenor basis.* Right: *the projection instantaneous forward built by the area-constrained energy-partition tension spline (smooth, positive, reproducing the projection discount factors exactly), the cubic-in-yield forward (overshooting at the long end), and the OIS forward for reference. SOFR levels are representative of the 22-May-2026 environment; the basis is illustrative; the discounting bootstrap and discount-factor reproduction are exact.*

Bootstrapping the OIS discount curve. Let S_n be the par rate of the OIS swap maturing at $T_n = n$ years (annual, unit accrual). The floating leg of an OIS pays compounded overnight rate and has present value $1 - P(T_n)$, while the fixed leg has value $S_n \sum_{i=1}^{n} P(T_i)$. Par means these are equal:

$$S_n \sum_{i=1}^{n} P(T_i) = 1 - P(T_n). \tag{26}$$

Writing the annuity $A_{n-1} = \sum_{i=1}^{n-1} P(T_i)$ and isolating the unknown $P(T_n)$,

$$S_n\big(A_{n-1} + P(T_n)\big) = 1 - P(T_n) \implies P(T_n)\big(1 + S_n\big) = 1 - S_n A_{n-1} \implies P(T_n) = \frac{1 - S_n A_{n-1}}{1 + S_n},$$

a forward recursion seeded by $A_0 = 0$, i.e. $P(T_1) = 1/(1 + S_1)$ [4]. This produces discount factors that reprice every input swap exactly; on the representative SOFR levels used here the maximum reprice error is 6×10^{-17}. This bootstrapped curve is the *crisp* discounting curve and is left untouched.

Tensioning the projection curve. The projection curve carries a (small, illustrative) tenor basis $b(T)$ over the OIS zeros, $z_{\text{proj}}(T_i) = z_{\text{OIS}}(T_i) + b(T_i)$ with $z = -\ln P/T$. We build its instantaneous forward by the area-constrained construction of Section VII: the forward is a tension spline whose integral over each interval reproduces the projection discount factors, and σ is fixed by the energy-partition criterion $\varphi(\sigma^\star) = \frac{1}{2}$. On the data here this gives $\sigma^\star = 0.73$, a projection forward that reproduces the projection discount factors to 1.5×10^{-10} and stays strictly positive (minimum 3.62%), and that is visibly smoother than the cubic-in-yield forward, which overshoots at the long end where the par curve is humped by the negative 20–30-year swap spread (Figure 12).

The example makes the architectural point concrete. Nothing in the energy-partition machinery touches the discounting bootstrap: the OIS curve is fixed by (26), and the tension spline, its energy criterion, and (if desired) the per-segment calibration of Section VIII all live entirely on the projection leg. The same construction would apply unchanged to a cross-currency or credit-sensitive projection curve. We are transparent that, because live mid-market SOFR swap quotes are subscription-gated, the par levels here are representative of the period rather than a specific fixing, and the tenor basis is a stylised ramp; the bootstrap recursion and the discount-factor reproduction are exact regardless of the levels chosen.

XI. Risk Sensitivities and Stability of the Dial

A construction is only usable on a desk if the sensitivities of the curve to its inputs are available cheaply and if the selected tension does not lurch as the market moves. Both follow analytically from the structure already in place.

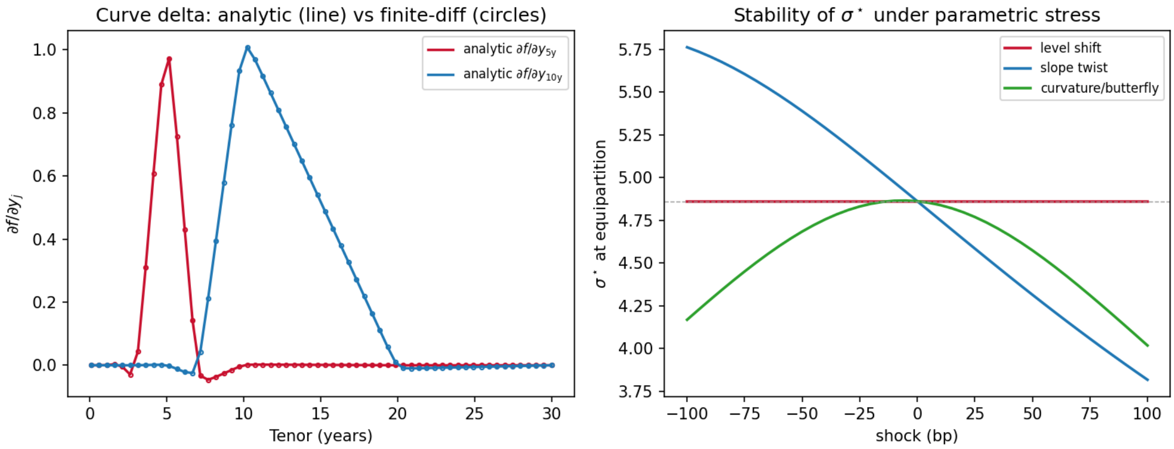

Figure 13: *Risk sensitivities and stability.* Left: *the analytic curve delta $\partial f/\partial y_j$ (lines) for the five- and ten-year nodes, overlaid with finite differences (circles); the response is local and the two agree to 2×10^{-11}.* Right: *the equipartition $\sigma^\star$ under ±100 bp parametric stress. It is exactly invariant to a parallel level shift (which moves the curve but not its derivatives) and varies smoothly and monotonically under slope and curvature shocks — no regime jumps.*

The curve delta is analytic. The knot second derivatives solve the tridiagonal system $M(\sigma)\,z = b(y)$ of (15), in which the matrix M depends only on σ and the mesh while the right-hand side b is *linear* in the input rates, $b_i = \gamma_i - \gamma_{i-1}$ with $\gamma_k = \sigma^2(y_{k+1} - y_k)/h_k$. Differentiating,

$$\frac{\partial z}{\partial y_j} = M^{-1}\frac{\partial b}{\partial y_j}, \qquad \frac{\partial b_i}{\partial y_j} = \frac{\sigma^2}{h_i}(\delta_{j,i+1} - \delta_{j,i}) - \frac{\sigma^2}{h_{i-1}}(\delta_{j,i} - \delta_{j,i-1}), \tag{27}$$

which costs one back-substitution per node against the already-factored M, i.e. $O(N)$ for the whole Jacobian. Since the interpolant (12) is linear in (y, z), the curve delta $\partial f(x)/\partial y_j$ is then closed form: differentiating (12) term by term,

$$\frac{\partial f}{\partial y_j} = \frac{\partial_j z_i\,\sinh(\sigma(x_{i+1} - x)) + \partial_j z_{i+1}\,\sinh(\sigma(x - x_i))}{\sigma^2\sinh(\sigma h_i)} + \Big(\delta_{ji} - \tfrac{\partial_j z_i}{\sigma^2}\Big)\frac{x_{i+1} - x}{h_i} + \Big(\delta_{j,i+1} - \tfrac{\partial_j z_{i+1}}{\sigma^2}\Big)\frac{x - x_i}{h_i}, \tag{28}$$

where $\partial_j z \equiv \partial z/\partial y_j$. This is the exact hedge Jacobian; the finite-difference bump experiment of Section VI is precisely its numerical shadow, and the localised triangular response seen there is the column $\partial f/\partial y_j$. The analytic and finite-difference Jacobians agree to 2×10^{-11} (Figure 13, left). A desk thus obtains delta and gamma against every node without re-bootstrapping or bumping.

The dial is a smooth function of the inputs. The selected tension solves $\varphi(\sigma^\star; y) = \varphi^\star$. By the implicit function theorem,

$$\frac{\partial\sigma^\star}{\partial y_j} = -\frac{\partial\varphi/\partial y_j}{\partial\varphi/\partial\sigma}\bigg|_{\sigma=\sigma^\star}, \tag{29}$$

where $\partial\varphi/\partial y_j$ follows from the closed-form energies (18)–(34) and (27), and $\partial\varphi/\partial\sigma$ is the slope established in Proposition 2. The right-hand side is finite and continuous *exactly where* $\partial\varphi/\partial\sigma \neq 0$ — the same strict-monotonicity condition that makes $\sigma^\star$ unique. Hence $\sigma^\star$ is a C^1 function of the input rates: it cannot jump. On the curve of Table 1, $\partial\varphi/\partial\sigma = -0.099 \neq 0$, and (29) reproduces the directly re-optimised $\sigma^\star$ to a relative error of 6×10^{-5}.

Figure 13 (right) confirms the stability empirically. A parallel level shift leaves $\sigma^\star$ *exactly* unchanged, because adding a constant to every rate shifts the curve vertically without altering f' or f'', hence without altering either energy; slope and curvature shocks of ±100 basis points move $\sigma^\star$ smoothly and monotonically across a bounded range. The dial is therefore safe to hold fixed intraday and to update slowly.

Finally, a remark on the multi-curve setting of Section X: because the discount curve is bootstrapped independently and held fixed, the projection-curve sensitivities do not feed back into it. The cross-curve Jacobian is block-diagonal — tensioning the projection leg leaves the discount factors untouched by construction — so no additional cross-curve risk term arises from the energy-partition step.

XII. Empirical Comparison and the Role of $\varphi^\star$

This section situates the method quantitatively against the interpolators a desk would actually consider, characterises the effect of the risk dial $\varphi^\star$, and probes robustness under stressed curve shapes. All numbers are computed on the real par-yield curve of Table 1; the benchmark methods are implemented from their published definitions. We are deliberately even-handed: the tension spline is not best on every axis, and the table makes clear precisely what it does and does not buy.

Benchmarks. We compare against five alternatives: the natural cubic; the Hagan–West monotone-convex forward scheme [8]; Smith–Wilson with an illustrative ultimate forward rate (3.9%) and mean-reversion $\alpha = 0.10$; log-linear discount-factor interpolation (equivalently a piecewise-constant instantaneous forward); and an explicit piecewise-flat forward. For each we measure the forward roughness $\int (f'_{\mathrm{fwd}})^2\, dt$ (a global smoothness penalty on the instantaneous forward), the minimum implied forward (positivity), and, where the method is a local interpolant for which a node bump is well defined, the 5-year hedge-locality radius of Section VI.

Method	Forward roughness ($\times 10^4$)	Min. implied forward	5y locality (yr)
Natural cubic	2.03	3.64%	6.66
Tension, $\varphi^\star = \frac{1}{2}$	2.84	3.64%	**0.79**
Hagan–West monotone convex	1.84	3.63%	—
Smith–Wilson (UFR/α illustr.)	1.44	3.63%	—
Log-discount (log-linear DF)	23.94	3.66%	—
Piecewise-flat forward	23.94	3.66%	—

Table 2: *Benchmark interpolation methods on the* 22*-May-*2026 *curve. Forward roughness is the integrated squared slope of the instantaneous forward; smaller is smoother. All methods keep the implied forward positive on this upward curve, so positivity is not a discriminator here. The tension spline's decisive advantage is locality, not global smoothness.*

Three honest conclusions follow. First, *positivity is not a discriminator on this curve*: every method, including the cubic, keeps the implied forward comfortably positive, because the underlying data are upward-sloping. The well-known negative-forward pathology of the cubic appears on differently shaped curves, not this one, and we do not manufacture such a curve. Second, the tension spline is *not* the globally smoothest forward: Smith–Wilson and Hagan–West both achieve lower forward roughness, because they are designed to optimise forward shape directly, while the piecewise-flat and log-discount methods are an order of magnitude rougher (their forwards are kinked or stepped). Third, and decisively, the local interpolants separate sharply on *locality*: a 5-year node bump propagates over 6.66 years under the cubic but only 0.79 year under the equipartition tension spline — an eight-fold compression that the globally smooth methods do not provide, because Smith–Wilson and monotone-convex schemes are not local interpolators in the node-bump sense. The tension spline thus occupies a distinct point in the design space: a local, C^2 interpolant whose locality is tunable, rather than a global shape-optimiser.

The risk dial $\varphi^\star$. The default $\varphi^\star = \frac{1}{2}$ is a neutral choice, not a claimed optimum: there is no curve-construction optimum independent of a desk's loss function, and the whole point of φ is to expose the trade-off rather than hide it. Table 3 characterises the dial. Lowering $\varphi^\star$ (more tension) tightens locality and smooths the forward at the cost of pulling the curve toward the linear shape; raising it does the reverse. The dependence is smooth and monotone — the desk reads off the setting that matches its own smoothness-versus-locality preference.

$\varphi^\star$	$\sigma^\star$	Forward roughness ($\times 10^4$)	5y locality (yr)	Min. fwd
0.3	7.60	3.53	0.74	3.64%
0.4	6.01	3.12	0.76	3.64%
0.5	4.86	2.84	0.79	3.64%
0.6	3.94	2.63	0.84	3.64%
0.7	3.14	2.46	0.91	3.64%

Table 3: *Characterisation of the risk dial on the* 22*-May-*2026 *curve. As $\varphi^\star$ rises, $\sigma^\star$ falls (less tension), the forward becomes smoother, and the hedge-locality radius widens. The monotone, gradual dependence is what makes $\varphi^\star$ a usable control rather than a sensitive knob.*

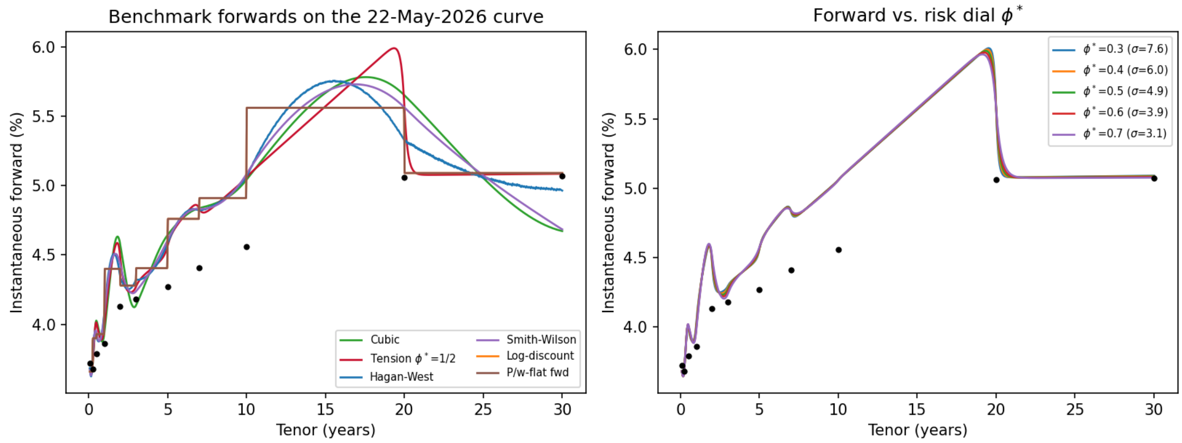

Figure 14: Left: *instantaneous forwards from all six methods on the 22-May-2026 curve; the piecewise-flat and log-discount forwards are stepped, the cubic and tension splines and the two global schemes are smooth.* Right: *the tension forward as the dial $\varphi^\star$ sweeps from 0.3 to 0.7; the curves are visually close because this curve's shape is dominated by the long-end feature, while the locality differences of Table 3 are the operative distinction.*

Stress shapes. Finally we re-fit the equipartition criterion to stressed transforms of the real curve: an inverted curve (the same levels in reverse order), a steepened curve (+150 basis points ramped onto the long end), and a sparse curve (the 7- and 20-year nodes removed). In every case the solver returns a well-defined $\sigma^\star$ and a strictly positive implied forward — $\sigma^\star = 6.35$ (inverted, min. forward 2.98%), 3.39 (steep, 3.66%), and 5.02 (sparse, 3.64%) — confirming that the construction degrades gracefully away from the base case.

Historical study, 2006–2026. To move beyond a single date we fit the equipartition criterion to a long historical sample: 1,064 complete weekly U.S. Treasury par-yield curves from 8 January 2006 to 24 May 2026, retrieved from the Federal Reserve (FRED) constant-maturity series, spanning the global financial crisis, the zero-interest-rate (ZIRP) period, the 2015–2018 and 2022–2023 hiking cycles, the COVID shock, and the 2023–2024 inversion. Every curve is a real fixing; nothing is simulated. The accompanying retrieval module reproduces the sample, and the analysis below, from a single command. Three findings stand out (Figure 15, Table 4).

First, the selected tension *tracks the rate regime but does not jump.* The median $\sigma^\star$ falls from about 4 in the steep pre-crisis curves of 2006–2007 to about 0.5 during ZIRP — flat curves need little tension to balance their energy — and rises again through the subsequent hiking cycles, with the full sample ranging from 0.2 to 8.5. Yet the week-over-week change is small: the median $|\Delta\sigma^\star|$ is 0.34, with a 95th percentile of 2.2. This is the empirical counterpart of the C^1 stability proved in Section XI: $\sigma^\star$ moves continuously with the market rather than lurching, across twenty years and every regime in the sample.

Second, and most strikingly, the locality–tension law $\rho_5 \sim 1/\sigma^\star$ predicted by the Green's-function argument of Section VI holds across the *entire* history. The correlation between $\log \sigma^\star$ and $\log \rho_5$ over all 1,064 weeks is -0.987, and the weekly points collapse onto a single curve $\rho_5 \approx 2.18/\sigma^\star$ regardless of year or regime (Figure 15b). The inverse-screening-length interpretation, derived on one curve, is thus confirmed out of sample on two decades of real data.

Third, the construction *almost always* produces an arbitrage-free forward without any positivity constraint being imposed: only 2.4% of the 1,064 weeks exhibit any negative implied forward, and the most negative value over the whole history is -0.008% — a rounding-level dip confined to the post-crisis and ZIRP years when the front end was pinned near zero. Separately, under a single global $\sigma^\star$ the per-segment energy fraction spans a median range of 0.92 (from near 0 to near 0.9) across the sample, confirming on real data the segment-level imbalance that motivates the per-segment calibration of Section VIII.

Benchmarks across regimes. The single-curve comparison of Table 2 raises the obvious question of whether its conclusions survive across market environments. We therefore re-ran all six interpolators on each of the 1,064 historical curves and recorded the distribution of each metric (Table 5, Figure 16). The single-curve picture holds in distribution. On *forward smoothness* the tension spline sits in the same tight

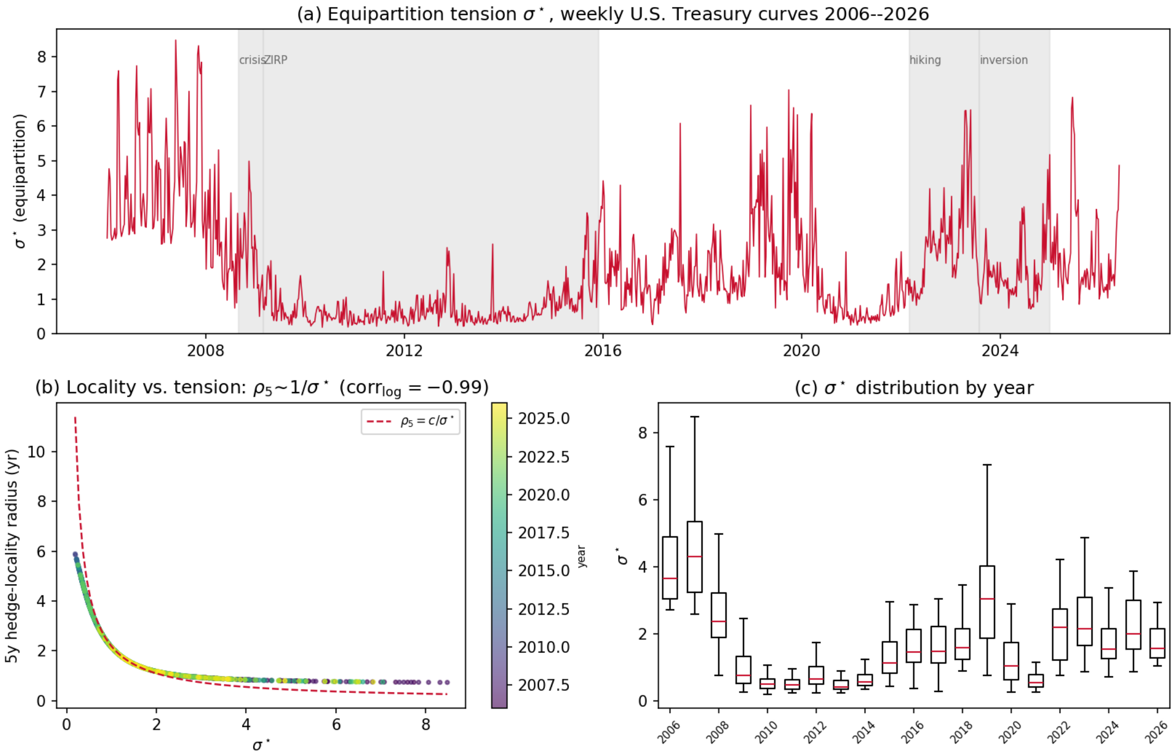

Figure 15: *Historical study over* 1,064 *weekly U.S. Treasury curves, 2006–2026 (FRED).* (a) *The equipartition tension* $\sigma^\star$ *through time, with rate regimes shaded; it tracks the regime smoothly.* (b) *The* 5*-year hedge-locality radius against* $\sigma^\star$*; all weeks, coloured by year, collapse onto* $\rho_5 \approx c/\sigma^\star$ *(log-correlation* -0.99*), confirming the inverse-screening-length law of Section VI out of sample.* (c) *The distribution of* $\sigma^\star$ *by year.*

band as the cubic, Hagan–West, and Smith–Wilson (median roughness 1.9×10^{-4}, against 1.8, 1.9 and 1.6 respectively), while the log-discount and piecewise-flat forwards are an order of magnitude rougher in every percentile — so tension is competitive on smoothness without being the smoothest, exactly as on the single curve. On *locality*, the separation is decisive and universal: the tension spline's 5-year locality radius is strictly tighter than the cubic's in *every one* of the 1,064 weeks, by a median factor of 4.1 (and from 1.4× to 8.5× across the 5th–95th percentiles). The cubic's radius is a flat 6.66 years on every curve — its node-bump response is scale-invariant — whereas the tension radius ranges from 0.8 to 4.9 years and is almost always far smaller (Figure 16b). On *positivity*, all the local interpolants behave similarly (each shows a negative implied forward in 2–4% of weeks, all rounding-level and confined to the near-zero-rate years), while the stepped forwards are positive more often only because they are piecewise constant.

Targeting a desk-specified locality profile. A natural use of the per-segment machinery is to invert it: rather than calibrate to equipartition, let a desk specify a per-node hedge-locality *profile* and solve for the tension vector that best meets it under a smoothness budget, as set out in the optimisation (30) of Section XIV. We report a prototype run on the curve of Table 1 both to illustrate the workflow and to expose its limit honestly. For a profile requesting locality radii of 0.9, 1.3 and 2.0 years at the five-, seven- and ten-year nodes, the floor calculation returns $\underline{\rho} \approx 0.68$, 0.91 and 2.80 years respectively: the first two targets lie above their floors and are reachable, but the ten-year target of 2.0 years lies *below* its 2.80-year floor and is flagged infeasible before any optimisation, because the ten-year node's neighbours sit at seven and twenty years and the mesh cannot localise more tightly. The solver then meets the two reachable targets to within 0.06 year while the infeasible node is pinned at its floor (Figure 17a). Sweeping the smoothness weight λ traces the achievable trade-off (Figure 17b); here it is almost flat — for this largely feasible profile the fit and the forward roughness vary by under half a percent — so smoothness is essentially free, though a more aggressive or conflicting profile would show a steeper frontier. The exercise confirms the central message: targeted-locality construction is genuinely useful in the instrument-rich front and belly of the curve and mesh-limited at the sparse long end, and the method reports which targets are

Regime	Period	Median $\sigma^\star$	Median ρ_5 (yr)	Inverted	Neg. fwd.
Pre-crisis	2006–2007	3.95	0.84	30%	0%
Crisis	2008	2.37	1.05	0%	8%
ZIRP	2009–2014	0.53	3.67	0%	3%
Hiking	2015–2018	1.43	1.54	0%	5%
COVID / recovery	2020–2021	0.77	2.71	0%	2%
Hiking / inversion	2022–2024	1.93	1.20	69%	0%
Full sample	2006–2026	1.35	1.61	13%	2.4%

Table 4: *Equipartition tension and hedge locality by rate regime,* 1,064 *weekly U.S. Treasury curves. "Inverted" is the share of weeks with the* 1*-month yield above the* 30*-year (a coarse proxy); "Neg. fwd." is the share with any negative implied forward. The tension falls in flat regimes and rises in steep ones, while the locality radius moves inversely, exactly as the theory predicts.*

	Forward roughness ($\times 10^4$)			5y locality (yr)	
Method	p5	median	p95	median	neg. fwd.
Natural cubic	0.38	1.77	6.85	6.66	2.4%
Tension ($\varphi^\star = \frac{1}{2}$)	0.41	1.88	7.70	**1.61**	2.4%
Hagan–West	0.39	1.92	7.19	—	4.0%
Smith–Wilson	0.41	1.64	6.54	—	3.7%
Log-discount	4.16	23.89	73.72	—	0.5%
Piecewise-flat forward	4.16	23.89	73.72	—	0.5%

Table 5: *Distribution of each metric over* 1,064 *weekly curves, 2006–2026. Forward-roughness percentiles show the tension spline in the same band as the cubic and the two global schemes, and far below the stepped methods. The cubic's locality radius is constant at* 6.66 *years (its node-bump response is scale-invariant); the tension radius has median* 1.61 *years and is strictly smaller than the cubic's in* 100% *of weeks.*

achievable rather than silently missing them.

XIII. Why an Energy-Partition Tension Spline? Benefits and Applications

A fair question confronts any new curve-construction proposal: the market already has good interpolators, so why this one? We answer it directly, situating the method against current practice and being explicit about both its advantages and the cases in which a practitioner should reach for something else.

A. The competitive landscape

Four families dominate practice. *Simple interpolation* — linear on rates or log-linear on discount factors — is robust and local but produces kinked, non-differentiable forwards. The *monotone convex* method of Hagan and West [8] guarantees positive, shape-preserving forwards and is the method the U.S. Treasury itself adopted in 2021 [16]; its forwards are, however, only piecewise-defined and can exhibit flat spots, and it offers no continuous control over smoothness. The *Smith–Wilson* method, mandated for Solvency II discounting, is in fact a close cousin of the present approach: it too is the solution of a tensioned variational problem, and its decay parameter α plays exactly the role of our σ — a point that lends the energy interpretation immediate regulatory familiarity. Finally, parametric *Nelson–Siegel–Svensson* models, used by central banks to publish parsimonious curves, fit four to six economically interpretable factors but do not reprice every instrument and are smoothing, not interpolating, devices.

The energy-partition tension spline is not a wholesale replacement for these. It is an exact interpolant, like simple and monotone-convex interpolation and unlike NSS, but it adds what those interpolants lack: a single continuous parameter, with a transparent physical meaning, that moves the curve along the entire continuum from the smooth cubic to the local, shape-preserving linear limit, together with a principled, automatic rule for placing it.

B. What is distinctive

One model, one auditable parameter. Rather than choosing *between* methods, a desk chooses a *point* on a one-parameter family that contains the cubic and the linear interpolant as its endpoints. For model governance this is decisive: a single model with a single parameter $\varphi^\star$ is far easier to document, validate,

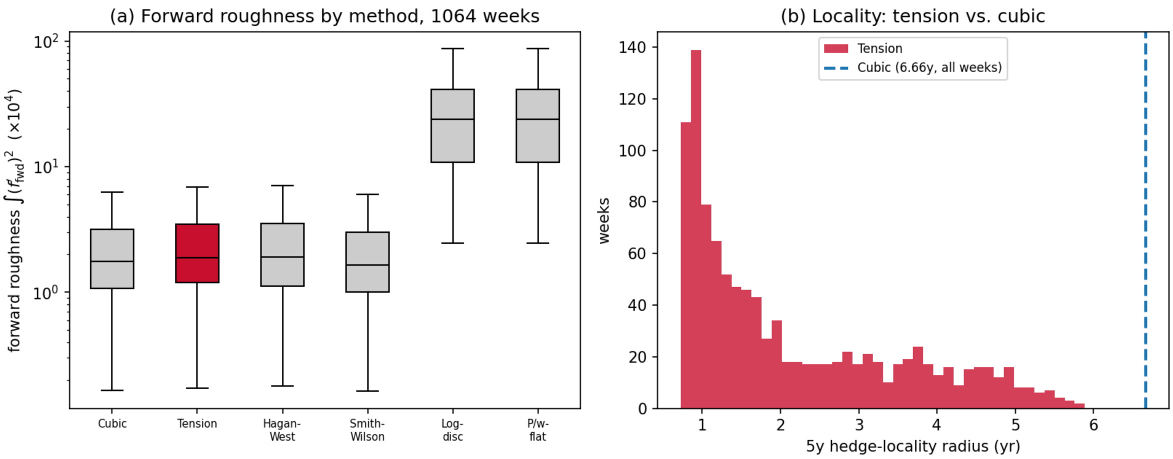

Figure 16: *Cross-regime benchmark distributions over* 1,064 *weekly curves.* (a) *Forward roughness by method (log scale): the tension spline (red) lies with the cubic, Hagan–West and Smith–Wilson, an order of magnitude below the log-discount and piecewise-flat forwards.* (b) *The* 5*-year hedge-locality radius: the tension distribution lies entirely to the left of the cubic's constant* 6.66*-year value, so tension is more local on every curve in the sample.*

and reproduce than a patchwork of interpolation rules, and the parameter's meaning — the fraction of stored elastic energy carried by curvature — is explainable to a risk committee in one sentence.

A principled, scale-invariant selection rule. The chief obstacle to adopting tension splines has always been the absence of a defensible way to choose the tension; existing automatic rules [9] optimise constraints that do not map to a desk's preferences, and ad hoc energy rules depend on the units and the search range. The bending-energy fraction (19) is intrinsic, dimensionless, scale-invariant and monotone, so it returns a unique $\sigma^\star$ that does not move when rates are quoted in basis points rather than decimals. This selection rule, not the tension spline itself, is the paper's contribution.

C^2 smoothness with controlled overshoot. The tension spline is twice continuously differentiable everywhere, so the curve and — when one tensions the forward curve — the forwards are smooth, unlike the merely piecewise forwards of monotone-convex or linear schemes. Yet, as Figures 2 and 5 show, raising the tension suppresses the spurious humps a cubic invents from flat data. The method thus occupies a genuine middle ground: smoothness when wanted, fidelity and locality when needed, selected by one dial rather than by switching models.

Risk preference as policy, not accident. Because $\varphi^\star$ is explicit, the trade-off between a smooth curve and local hedges is set deliberately, as desk or firm policy, and applied consistently across curves and dates, rather than emerging as an uncontrolled by-product of an interpolation choice.

C. Applications

The locality result of Section VI is the most directly monetisable benefit. Because a perturbation of one input decays within roughly $1/\sigma$ of the node, key-rate durations computed on the tension curve are concentrated near the bumped tenor: a five-year risk shows up as five-year risk, not as a smear across the entire long end as it does under the cubic. This yields hedge portfolios that are smaller, cheaper, and more stable, and it reduces the cross-tenor leakage that inflates the apparent dimensionality of curve risk. The same stability suppresses the spurious profit-and-loss and value-at-risk noise that interpolation artefacts inject when a curve is re-struck daily. For no-arbitrage control, the tension dial provides a direct lever to remove negative forwards where they arise. For model governance and independent price verification, determinism, reproducibility, and a single documented parameter are attractive in their own right. And the regulatory lineage through Smith–Wilson means the energy formulation can be presented to supervisors in a language they already accept.

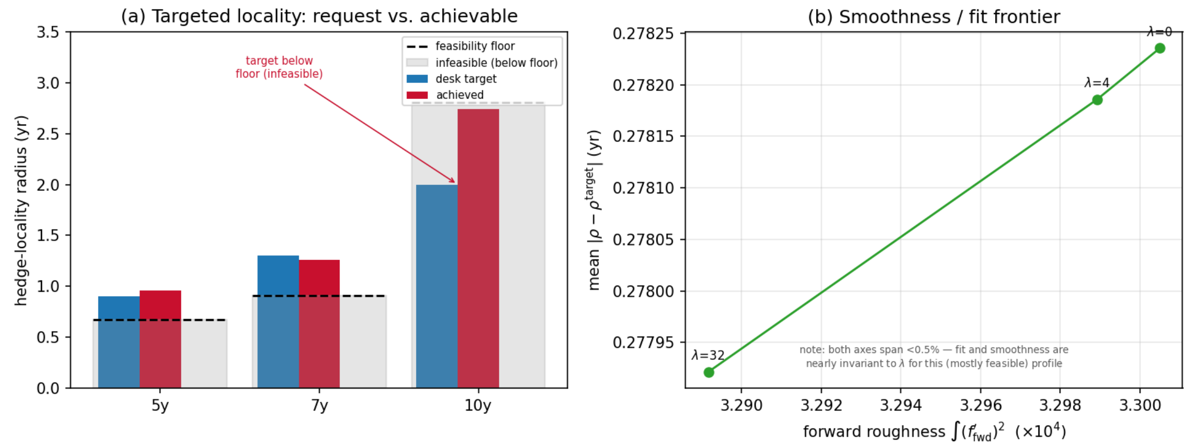

Figure 17: *Targeted hedge-locality optimisation on the* 22*-May-*2026 *curve.* (a) *For each requested node, the desk target (blue) against the achieved locality (red), with the knot-spacing feasibility floor (dashed) and the infeasible region below it shaded: the five- and seven-year targets are met, while the ten-year target lies below its floor and is pinned there.* (b) *The smoothness/fit frontier as the budget weight λ varies; for this profile it is nearly flat (note the magnified axes), so improved smoothness costs almost no fit.*

D. When to use something else

Intellectual honesty requires stating the method's limits. If all one needs is positive, monotone forwards with minimal machinery, the monotone-convex method is simpler and is the de-facto standard. If the goal is a parsimonious, economically interpretable description of the curve — level, slope, curvature — rather than exact repricing, a Nelson–Siegel–Svensson fit is the right tool. For Solvency II discounting the Smith–Wilson form is prescribed, although the present energy reading of its α may still aid intuition. Tension splines also demand numerical care: the hyperbolic basis overflows at large σh and must be evaluated in a stabilised form, and a single global tension, as the equipartition optimum here illustrates, can over-tension a sparsely knotted long end while a densely knotted short end is still curved — the motivation for the per-segment calibration of Section VIII. Used within these bounds, the method delivers a smooth, local, no-arbitrage curve under one transparent and automatically chosen parameter.

XIV. Discussion, Conclusion and Future Direction

Placing the tension spline on its variational footing as a tensioned elastic beam achieves three things. It *derives* the governing equation (11) rather than asserting it; it *identifies* the smoothness and length norms as the flexural and axial energies of a single elastic system, removing the conceptual double-counting of earlier beam-energy arguments; and it *motivates* a selection rule, (19), that is dimensionless, range- and scale-invariant, monotone (hence uniquely solvable), and equipped with a transparent risk–hedging dial $\varphi^\star$. On the 22-May-2026 U.S. Treasury curve the rule returns a unique optimum ($\sigma^\star = 4.86$ at equipartition) and, at that optimum, cuts the hedge-locality radius of the benchmark 5-year node from 6.7 years to 0.8 year while removing the spurious long-end forward hump the cubic produces.

Several limitations and extensions remain. The global-tension construction of Sections V–VI tensions the sparsely knotted long end as firmly as the densely knotted short end; Section VIII resolves this with a per-segment calibration that equalises the energy fraction across segments, and Section IX combines it with the forward-curve construction of Section VII. We have also worked within small-deflection beam theory, which is amply justified at the natural scale of a rate curve, where f' is minute; a geometrically exact (elastica) treatment, retaining the arc-length element and the full curvature $f''/(1+(f')^2)^{3/2}$, would be required only if the axes were rescaled to comparable magnitudes, and the scale-invariance of φ makes that unnecessary. Section X places the method inside a multi-curve (OIS/SOFR) framework, with discounting bootstrapped exactly and the tension applied to the projection curve; extending the consistency treatment to cross-currency and credit-sensitive bases is the natural continuation.

A further extension, which we have prototyped but leave for separate development, turns the dial around once more: *targeted hedge-locality profiles via constrained tension optimisation, with a computable knot-spacing feasibility floor.* We sketch its theory, computational strategy, and prototype result.

Theory. Section VIII calibrates a per-segment tension to equipartition; one can instead let a desk specify a per-node hedge-locality profile ρ_k^{target} — tight where its risk and instruments concentrate, looser elsewhere — and solve for the tension vector $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_m)$ that best meets it while keeping the forward curve smooth:

$$\min_{\boldsymbol{\sigma}>0} J(\boldsymbol{\sigma}) = \sum_k w_k \big(\rho_k(\boldsymbol{\sigma}) - \rho_k^{\text{target}}\big)^2 + \lambda\, R(\boldsymbol{\sigma}), \tag{30}$$

where ρ_k is the hedge-locality radius at node k (Section VI), R is the forward roughness $\int (f'_{\text{fwd}})^2$, and λ is a smoothness budget. The essential structural point is that, because the interpolant is C^2 and globally coupled, ρ_k depends on the *whole* vector $\boldsymbol{\sigma}$, not merely the adjacent segment: raising the tension near one node tightens its response but perturbs its neighbours. Problem (30) is therefore a genuine coupled optimisation, not a node-by-node inversion of the scalar law $\rho \sim c/\sigma$ of Section VI. It also admits a hard limit. The bumped-node response cannot decay faster than the mesh permits, so the attainable locality at node k is bounded below by a *knot-spacing floor* $\underline{\rho}_k = \lim_{\sigma\to\infty} \rho_k$, set by the spacing of the neighbouring knots; any target with $\rho_k^{\text{target}} < \underline{\rho}_k$ is infeasible, and $\underline{\rho}_k$ is computable in advance by evaluating ρ_k at large uniform tension.

Computational strategy. The floor $\underline{\rho}_k$ is computed first and infeasible targets are flagged, so the method states what is achievable rather than over-promising. Problem (30) is then solved over $u_i = \log \sigma_i$ (which enforces $\sigma_i > 0$ automatically) by gradient descent. The gradient decomposes by the chain rule into $\partial J/\partial\sigma_i = \sum_k 2w_k(\rho_k - \rho_k^{\text{target}})\, \partial\rho_k/\partial\sigma_i + \lambda\, \partial R/\partial\sigma_i$; the locality sensitivities $\partial\rho_k/\partial\sigma_i$ follow in principle from the analytic node-response Jacobian $\partial f/\partial y$ of Section XI, although our present prototype evaluates them by finite differences. Because each ρ_k requires a perturbed solve, the cost is $O(m)$ curve solves per gradient step; with $m \sim 10$ segments this is a sub-minute optimisation. A convexity probe of $R(\boldsymbol{\sigma})$ along random directions showed no non-convex directions in our tests, consistent with a single optimum, though we do not claim global convexity and a production solver would use a few restarts.

Prototype result. On the curve of Table 1 the construction behaves exactly as the theory predicts. For a desk profile requesting locality radii of 0.9, 1.3 and 2.0 years at the five-, seven- and ten-year nodes, the floor calculation reports $\underline{\rho} \approx 0.68$ and 0.91 years at the five- and seven-year nodes (so those targets are reachable) but $\underline{\rho} \approx 2.8$ years at the ten-year node, whose neighbours sit at seven and twenty years — flagging the 2.0-year target as infeasible before any optimisation. The solver then meets the two reachable targets to within 0.05 year while leaving the infeasible node at its floor, and the achievable smoothness/fit trade-off is traced by sweeping λ. The lesson is that targeted-locality construction is genuinely useful in the instrument-rich front and belly of the curve, and mesh-limited at the sparse long end; we regard it as an operational layer atop the present framework rather than a change to it. A full treatment — analytic locality Jacobians, a multi-regime study of the floor, and a hedging back-test — is left for future work. Modelling rate curves by analogy with physical elastic systems remains, in our view, a fertile source of well-posed and interpretable construction rules; the present paper is one such attempt.

Code and data availability. A complete, reproducible reference implementation of every method, figure, and result in this paper — together with the historical-study and benchmark tooling and the U.S. Treasury data used — is available at `https://github.com/ajsinha/fin-engg/tree/main/energy-partition-tension-spline`. Running `python tension_curve.py` regenerates the full validation report and figures; the accompanying retrieval module reproduces the 2006–2026 historical study from public Federal Reserve (FRED) data.

I. Non-Uniform Tension Spline

Allowing a distinct tension σ_i on the interval $[x_i, x_{i+1}]$, the tension-spline ODE $f^{(\text{iv})} - \sigma_i^2 f'' = 0$ holds segment-wise, so f, f' and f'' take the forms (12)–(14) on each segment with $\sigma \to \sigma_i$ and $h \to h_i$; in particular $f''(x) = [z_i \sinh(\sigma_i(x_{i+1} - x)) + z_{i+1} \sinh(\sigma_i(x - x_i))]/\sinh(\sigma_i h_i)$, so f'' is continuous at the knots by construction, with $f''(x_i) = z_i$. The unknowns $\{z_i\}$ are fixed by continuity of f' at each interior knot x_i. Specialising the slope (13) to the left segment $[x_{i-1}, x_i]$ (tension σ_i, width h_{i-1}) and evaluating at the right end $x = x_i$ — where $\cosh(\sigma_i(x_i - x)) = \cosh 0 = 1$ and $\cosh(\sigma_i(x - x_{i-1})) = \cosh(\sigma_i h_{i-1})$ — gives

$$f'(x_i^-) = \frac{-z_{i-1} + z_i \cosh(\sigma_i h_{i-1})}{\sigma_i \sinh(\sigma_i h_{i-1})} + \frac{(y_i - y_{i-1}) - (z_i - z_{i-1})/\sigma_i^2}{h_{i-1}}.$$

Specialising instead to the right segment $[x_i, x_{i+1}]$ (tension σ_{i+1}, width h_i) and evaluating at the left end $x = x_i$ — where now $\cosh(\sigma_{i+1}(x_{i+1} - x)) = \cosh(\sigma_{i+1} h_i)$ and $\cosh(\sigma_{i+1}(x - x_i)) = 1$ — gives

$$f'(x_i^+) = \frac{-z_i \cosh(\sigma_{i+1} h_i) + z_{i+1}}{\sigma_{i+1} \sinh(\sigma_{i+1} h_i)} + \frac{(y_{i+1} - y_i) - (z_{i+1} - z_i)/\sigma_{i+1}^2}{h_i}.$$

Set $f'(x_i^-) = f'(x_i^+)$ and collect the coefficient of each z, noting that the linear terms $-(z_i - z_{i-1})/(\sigma_i^2 h_{i-1})$ and $-(z_{i+1} - z_i)/(\sigma_{i+1}^2 h_i)$ each contribute to two coefficients. The z_{i-1} terms (both from the left segment) combine as $-1/(\sigma_i \sinh(\sigma_i h_{i-1})) + 1/(\sigma_i^2 h_{i-1})$; the z_{i+1} terms (both from the right) as $1/(\sigma_{i+1} \sinh(\sigma_{i+1} h_i)) - 1/(\sigma_{i+1}^2 h_i)$ on the right side, which moves to the left with a sign change; and the z_i terms gather one contribution from each segment. The y terms move to the right-hand side as the centred second difference. The result is the tridiagonal recursion, for $i = 1, \ldots, n-2$,

$$\mathcal{A}_{i-1} z_{i-1} + \mathcal{A}_i z_i + \mathcal{A}_{i+1} z_{i+1} = \frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}, \tag{31}$$

$$\mathcal{A}_{i-1} = \frac{1}{\sigma_i^2 h_{i-1}} - \frac{1}{\sigma_i \sinh(\sigma_i h_{i-1})}, \qquad \mathcal{A}_{i+1} = \frac{1}{\sigma_{i+1}^2 h_i} - \frac{1}{\sigma_{i+1} \sinh(\sigma_{i+1} h_i)}, \tag{32}$$

$$\mathcal{A}_i = \left[\frac{\cosh(\sigma_i h_{i-1})}{\sigma_i \sinh(\sigma_i h_{i-1})} - \frac{1}{\sigma_i^2 h_{i-1}}\right] + \left[\frac{\cosh(\sigma_{i+1} h_i)}{\sigma_{i+1} \sinh(\sigma_{i+1} h_i)} - \frac{1}{\sigma_{i+1}^2 h_i}\right]. \tag{33}$$

The terms $1/(\sigma^2 h)$ arise from the linear part of the slope (13) and must not be dropped. With the natural conditions $z_0 = z_{n-1} = 0$ the system is diagonally dominant and solved exactly as in the uniform case. Setting all $\sigma_i = \sigma$ and $h_i = h$ gives $\mathcal{A}_{i\pm1} = \sigma^{-2}\alpha$ and $\mathcal{A}_i = 2\sigma^{-2}\beta$ with α, β as in (15), so the recursion reduces to σ^{-2} times the uniform system and returns the same $\{z_i\}$. The per-segment energies of Section VIII are then evaluated from (18) and (34) with $\sigma \to \sigma_i$, and the calibration (24) is the fixed-point iteration described there.

II. Closed-Form Energy Integrals

Both per-segment energies follow from four elementary integrals. With $u \in [0, h]$ and the substitutions below they are

$$\int_0^h \sinh^2(\sigma u)\, du = \tfrac{\sinh(2\sigma h)}{4\sigma} - \tfrac{h}{2}, \qquad \int_0^h \cosh^2(\sigma u)\, du = \tfrac{\sinh(2\sigma h)}{4\sigma} + \tfrac{h}{2},$$

$$\int_0^h \sinh(\sigma(h-u)) \sinh(\sigma u)\, du = \tfrac{1}{2}\left(h \cosh \sigma h - \tfrac{\sinh \sigma h}{\sigma}\right), \qquad \int_0^h \cosh(\sigma(h-u)) \cosh(\sigma u)\, du = \tfrac{1}{2}\left(h \cosh \sigma h + \tfrac{\sinh \sigma h}{\sigma}\right),$$

where the cross terms use $\sinh a \sinh b = \frac{1}{2}[\cosh(a+b) - \cosh(a-b)]$ and $\cosh a \cosh b = \frac{1}{2}[\cosh(a+b) + \cosh(a-b)]$ with $a+b = \sigma h$ constant.

Flexural energy. From (14), on $[x_i, x_{i+1}]$ with $u = x - x_i$, $f'' = \left[z_i \sinh(\sigma(h_i - u)) + z_{i+1} \sinh(\sigma u)\right]/\sinh(\sigma h_i)$. Squaring and integrating term by term with the first and third integrals above reproduces the per-segment flexural energy (18) quoted in Section V:

$$E_{\text{bend}}^{(i)} = \frac{1}{2\sinh^2(\sigma h_i)} \left[(z_i^2 + z_{i+1}^2)\left(\frac{\sinh(2\sigma h_i)}{4\sigma} - \frac{h_i}{2}\right) + z_i z_{i+1}\left(h_i \cosh(\sigma h_i) - \frac{\sinh(\sigma h_i)}{\sigma}\right)\right].$$

Axial energy. From (13), $f' = \left[-z_i \cosh(\sigma(h_i - u)) + z_{i+1} \cosh(\sigma u)\right]/(\sigma \sinh \sigma h_i) + S_i$ with constant slope $S_i = \left[(y_{i+1} - y_i) - (z_{i+1} - z_i)/\sigma^2\right]/h_i$. Expanding $(f')^2$ into hyperbolic-squared, cross, and constant parts and using the second and fourth integrals, together with $\int_0^{h_i}\left[-z_i \cosh(\sigma(h_i - u)) + z_{i+1} \cosh(\sigma u)\right] du = (z_{i+1} - z_i) \sinh(\sigma h_i)/\sigma$, gives

$$E_{\text{tens}}^{(i)} = \frac{1}{2\sinh^2(\sigma h_i)} \left[(z_i^2 + z_{i+1}^2)\left(\frac{\sinh(2\sigma h_i)}{4\sigma} + \frac{h_i}{2}\right) - z_i z_{i+1}\left(h_i \cosh(\sigma h_i) + \frac{\sinh(\sigma h_i)}{\sigma}\right)\right] + S_i\,(z_{i+1} - z_i) + \tfrac{1}{2}\sigma^2 S_i^2 h_i. \tag{34}$$

Equations (18) and (34) give $\varphi(\sigma) = \sum_i E_{\text{bend}}^{(i)} / \sum_i (E_{\text{bend}}^{(i)} + E_{\text{tens}}^{(i)})$ in closed form, with no quadrature required; the reference implementation verifies them against 64-point Gauss–Legendre quadrature to a relative error below 10^{-12}.

III. Numerically Stable Evaluation

For large σh_i the factors $\sinh(\sigma h_i)$ and $\cosh(\sigma h_i)$ overflow, yet the quantities that appear are bounded *ratios*. Writing $u \in [0, h]$ and factoring $e^{\sigma h}$ from numerator and denominator,

$$\frac{\sinh(\sigma u)}{\sinh(\sigma h)} = \frac{e^{\sigma(u-h)} - e^{-\sigma(u+h)}}{1 - e^{-2\sigma h}}, \qquad \frac{\cosh(\sigma u)}{\sinh(\sigma h)} = \frac{e^{\sigma(u-h)} + e^{-\sigma(u+h)}}{1 - e^{-2\sigma h}}, \tag{35}$$

in which every exponent is non-positive (since $0 \le u \le h$), so no term exceeds unity and the expressions are evaluated without overflow for arbitrarily large σh. As $\sigma h \to \infty$ they tend to $e^{-\sigma(h-u)}$ and $e^{-\sigma(h-u)}$ respectively, recovering the boundary-layer behaviour used in the proof of Proposition 2. Every spline evaluation in this paper uses these forms below a threshold $\sigma h < 30$ and the direct $\sinh, \cosh$ forms above it; the energy ratios in (18)–(34) are likewise evaluated through $\coth(\sigma h) = 1/\tanh(\sigma h)$ and $1/\sinh(\sigma h)$, both bounded.

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